PHYSICS GRADE 10: RADIOACTIVITY AND STABILITY OF ISOTOPES

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.1 (7 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Physics
Strand	Strand 3.0: Electricity and Magnetism
Sub-Strand	Sub-Strand 3.1: Radioactivity and Stability of Isotopes
Total Duration	7 lessons × 40 minutes = 280 minutes total
Sub-Strand Content	– Meaning of radioactivity and the structure of the nucleus (protons, neutrons, nucleon number, proton number) – Isotopes: definition, examples (e.g. Carbon-12 and Carbon-14), and the concept of nuclear stability – Neutron-to-proton (N/Z) ratio and construction of the belt (valley) of stability – Types of nuclear radiation: alpha (α), beta (β), and gamma (γ) — properties, charge, mass, and penetrating power – Detection of radiation: Geiger-Müller counter, cloud chamber, and photographic film — working principles and applications – Radioactive decay and half-life: decay curves, calculations, and real-world significance – Nuclear equations: writing and balancing equations for α-decay, β-decay, and γ-emission – Nuclear fission and nuclear fusion: energy release, comparison, and global/Kenyan energy relevance – Applications of radioactivity: medicine (cancer treatment, PET scans), agriculture (food irradiation, soil tracing), industry, and carbon dating – Radiation safety: effects of radiation on living tissue, protection measures, and responsible handling of radioactive materials
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) explain the concept of radioactivity and describe the structure of the nucleus in terms of protons and neutrons, b) define isotopes and distinguish between stable and unstable isotopes using the neutron-to-proton (N/Z) ratio and the belt of stability, c) describe the properties of alpha (α), beta (β), and gamma (γ) radiation in terms of charge, mass, speed, and penetrating power, d) explain how radiation detection devices (Geiger-Müller counter, cloud chamber) work and identify their applications in medicine and environmental monitoring in Kenya, e) interpret radioactive decay curves to determine half-life and perform calculations involving half-life, f) write and balance nuclear equations for alpha decay, beta decay, and gamma emission, g) distinguish between nuclear fission and nuclear fusion, explaining the energy changes involved and their relevance to global and Kenyan energy needs, h) evaluate the applications of radioactivity in medicine, agriculture, and industry within the Kenyan context, i) assess the health and safety risks of radiation exposure and propose appropriate protective measures, j) appreciate the role of nuclear science in solving real-world problems in Kenya and beyond.
Core Competencies	– Communication and Collaboration: Learners work in groups to compare α, β, and γ radiation properties, construct the belt of

	stability, and present findings on radiation applications — embracing teamwork and contributing to shared group decisions. – Critical Thinking and Problem Solving: Learners analyse neutron-to-proton ratios to predict nuclear stability, interpret decay curves to calculate half-life, and balance nuclear equations — developing logical reasoning and quantitative problem-solving skills. – Learning to Learn: Learners use PhET simulations, Khan Academy resources, and YouTube videos to independently investigate radioactive decay and half-life, practising self-directed inquiry and knowledge construction. – Digital Literacy: Learners interact with digital tools (PhET Nuclear Fission simulation, online decay curve plotters) to visualise abstract nuclear processes, building competency in using technology for scientific investigation. – Citizenship: Learners evaluate radiation safety protocols and the ethical use of nuclear technology in Kenyan hospitals and research institutions, developing informed and responsible civic perspectives on science and society.
Core Values	– Responsibility: Learners demonstrate responsibility by following radiation safety guidelines during simulations and discussions, and by considering the ethical implications of nuclear technology use in Kenyan medical and industrial contexts. – Respect: Learners show respect for diverse perspectives when discussing the benefits and risks of nuclear energy and radioactive applications, appreciating contributions from peers during group activities. – Unity: Learners collaborate across ability groups to construct stability belt diagrams and simulate decay curves, fostering a spirit of togetherness and shared learning. – Social Justice: Learners examine equitable access to nuclear medicine technologies (e.g., cancer treatment via radiotherapy) in Kenyan public hospitals, developing awareness of health inequalities and advocating for fair resource distribution.
Science & Engineering Practices	– Asking Questions and Defining Problems: Learners pose and refine investigable questions about why some nuclei are unstable and what determines the mode of decay, anchored in the puzzling observation that identical elements can behave very differently. – Developing and Using Models: Learners construct and iteratively revise a visual model of the belt of stability plotting N vs Z, and build cumulative models of nuclear decay processes across lessons. – Planning and Carrying Out Investigations: Learners design and carry out a simulated radioactive decay experiment (using dice or PhET) to collect data on decay rates and half-life, making decisions about trials and data recording. – Analysing and Interpreting Data: Learners plot and interpret radioactive decay curves, extracting half-life values and comparing theoretical predictions with simulated data. – Using Mathematics and Computational Thinking: Learners apply half-life equations ($N = N_0(½)^n$) and balance nuclear equations, using quantitative reasoning to solve decay problems in Kenyan agricultural and medical contexts. – Constructing Explanations: Learners use the N/Z ratio and binding energy concepts to construct evidence-based explanations for nuclear stability and the type of decay a nucleus undergoes. – Engaging in Argument from Evidence: Learners debate the benefits and risks of nuclear technology in Kenya (e.g., proposed nuclear energy, radiotherapy at KNH), supporting their positions with data and scientific reasoning. – Obtaining, Evaluating, and Communicating Information: Learners research and present on real-world applications of radioactivity in Kenyan medicine, agriculture (e.g., KARI/KALRO use of radiation in crop improvement), and industry, evaluating source credibility.
Pertinent & Contemporary Issues (PCIs)	– Health Education: Learners discuss the biological effects of ionising radiation on human tissue and the importance of radiation safety in medical facilities such as Kenyatta National Hospital (KNH) and Moi Teaching and Referral Hospital, promoting health-conscious decision-making. – Environmental Education: Learners examine how radioactive contamination can affect ecosystems (e.g., water sources like Lake Victoria and the Tana River) and discuss responsible waste disposal of radioactive materials. – Life Skills — Critical Thinking and Decision Making: Learners evaluate competing claims about nuclear energy as a solution to Kenya's energy challenges versus its safety and environmental risks, building informed decision-making skills.

	– Career Guidance: Learners identify and explore career pathways in nuclear medicine, radiography, radiation protection, and nuclear engineering — including opportunities at the Kenya Nuclear Electricity Board (KNEB) and local universities offering relevant programmes. – Disaster Risk Reduction: Learners assess the risks of radiation exposure in emergency scenarios and develop awareness of safety protocols, connecting to national disaster preparedness frameworks.
Career Connections	– Radiographer / Radiologist: This sub-strand directly explains the physics of radiation used in X-ray, CT, and PET imaging — equipping learners to pursue radiography training available at Kenyan universities such as University of Nairobi and KMTC. – Nuclear Medicine Specialist: Learners explore how radioactive isotopes (e.g., Iodine-131, Technetium-99m) are used in diagnosing and treating diseases, providing a foundation for medical careers with a nuclear science component at facilities like KNH. – Radiation Protection Officer: Understanding radiation types, penetration, and safety measures in this sub-strand prepares learners for roles in monitoring and controlling radiation exposure in hospitals, industries, and research institutions regulated by the Kenya Nuclear Regulatory Authority (KNRA). – Agricultural Scientist / Researcher: The application of radiation in crop mutation breeding and soil tracing, practiced at KALRO (Kenya Agricultural and Livestock Research Organisation), connects directly to agri-science career pathways. – Nuclear/Energy Engineer: The fission and fusion lesson connects to Kenya's exploration of nuclear energy through KNEB, inspiring learners to consider engineering careers contributing to Kenya's long-term energy security under Vision 2030. – Environmental Health Scientist: Learners who study radiation monitoring and contamination pathways are prepared for careers in environmental protection agencies and public health bodies in Kenya.
Focus for Lessons	This unit introduces Grade 10 learners to the fascinating and sometimes invisible world of nuclear physics — exploring why some atomic nuclei are inherently unstable, how they release energy through radioactive decay, and how humanity has harnessed this knowledge for medicine, agriculture, and energy. Using a phenomenon-driven, investigation-rich approach, learners move from puzzling over why carbon-14 behaves differently from carbon-12, to constructing the belt of stability, characterising the three types of radiation, simulating half-life decay, and evaluating the real-world applications and risks of nuclear technology in the Kenyan context. The unit is grounded in the NGSS Science and Engineering Practices, emphasising model-building, data analysis, and evidence-based argumentation throughout all seven lessons.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: "Why do some atoms in Kenyan rocks spontaneously release invisible energy — and how can we harness or protect ourselves from that energy?" KICD KEY INQUIRY QUESTIONS: 1. What makes some atomic nuclei stable while others are unstable and undergo radioactive decay? 2. How does knowledge of radioactivity help us solve problems in medicine, agriculture, energy, and environmental monitoring in Kenya?
Anchoring Phenomenon	ANCHORING PHENOMENON — "The Invisible Killer in the Rock": A sample of pitchblende ore (uranium ore) is placed near a photographic film in complete darkness. When the film is developed, it shows a clear exposure — something invisible has passed through the dark container and affected the film. No light, no heat, no sound. Yet the rock has left its mark. This is puzzling: rocks are assumed to be inert, stable objects. Why would a rock spontaneously release invisible energy? Where does that energy come from? Why do only some rocks do this? And why does it never seem to stop — the rock keeps emitting

	radiation day after day without any external energy input? Learners then discover that uranium ore contains atoms whose nuclei are unstable — their neutron-to-proton ratios fall outside the "belt of stability" — and that these nuclei spontaneously reorganise themselves by ejecting particles or energy (α, β, γ radiation) in an effort to reach a more stable configuration. The anchoring phenomenon threads through every lesson: each new concept (isotopes, decay types, detection, half-life, nuclear equations, fission/fusion, applications) is explicitly connected back to explaining what is happening inside that rock, at the nuclear level, and why it matters for Kenya and the world.
Storyline Thread	Lesson 1 — Terminologies, Nuclear Structure & Isotopes: Learners are confronted with the anchoring phenomenon (pitchblende film exposure). They predict what could cause it, then investigate the structure of the nucleus, define key terms (nucleon, nuclide, isotope, radioactivity), and examine why Carbon-14 and Carbon-12 — chemically identical — behave so differently. Groups construct N vs Z plots and are introduced to the belt of stability. The Driving Question Board (DQB) is opened and initial questions are posted. Lesson 2 — Types of Radiation (α, β, γ): Learners design a predict-observe-explain activity comparing the penetrating power of the three radiation types using paper, aluminium, and lead analogues (diagrams/simulation). They tabulate properties (charge, mass, speed, ionising ability, penetration) and connect each type back to which unstable nuclei emit it. Groups update their N vs Z stability model with decay arrows showing how α and β emissions shift a nucleus toward the belt of stability. Lesson 3 — Detection of Radiation: Learners observe a virtual Geiger-Müller counter (PhET / YouTube) and a cloud chamber simulation, explaining the working principle of each. They investigate background radiation and discuss how detection devices are used in Kenyan hospitals (KNH radiography) and environmental monitoring (KNRA). Learners revise their DQB — adding answers and new questions about measurement and safety. Lesson 4 — Half-Life and Decay Curves: Learners simulate radioactive decay using 100 six-sided dice (or the PhET "Nuclear Decay" simulation), rolling and removing "decayed" dice each round, plotting a decay curve. They extract the half-life graphically, apply the equation $N = N_0(½)$^n to solve problems, and connect half-life to carbon dating at Olorgesailie and Iodine-131 dosing at KNH. Decay curves are added to cumulative models. Lesson 5 — Nuclear Equations: Learners write and balance nuclear equations for α-decay (e.g., Uranium-238 → Thorium-234), β-decay (e.g., Carbon-14 → Nitrogen-14), and γ-emission, using conservation of nucleon number and proton number. They verify how each decay shifts the daughter nucleus on the N vs Z plot toward the belt of stability, directly connecting equations back to the anchoring phenomenon and the stability model built in Lesson 1. Lesson 6 — Nuclear Fission and Fusion: Learners use PhET "Nuclear Fission" simulation and a YouTube explainer to investigate how splitting a uranium nucleus (fission) or fusing hydrogen nuclei (fusion) releases enormous energy. They compare mass defect and binding energy conceptually, debate Kenya's nuclear energy prospects (KNEB context), and construct an evidence-based argument about whether Kenya should pursue nuclear power. The DQB is updated with new understanding about energy sources. Lesson 7 — Applications, Safety, and Unit Review: Learners synthesise all prior lessons by exploring applications of radioactivity (radiotherapy at KNH, KALRO crop irradiation, industrial thickness gauging, carbon dating) and evaluating radiation safety measures (ALARA principle, shielding, distance, time). Groups present a final revised cumulative model — from the original puzzling rock phenomenon to a full nuclear physics explanatory framework. The DQB is formally closed, with all opening questions answered or flagged for further study. A structured review consolidates key concepts and learning outcomes.

Supporting Phenomena	– Carbon-14 Dating of Ancient Kenyan Artefacts: Archaeologists at sites like Olorgesailie (Rift Valley) use carbon-14 dating to determine the age of ancient tools and bones. Learners explore: why does carbon-14 decay but carbon-12 remain stable? This supports understanding of isotopes and the belt of stability. – Radiation Therapy at Kenyatta National Hospital: Cancer patients in Nairobi receive targeted gamma radiation to destroy tumour cells. Learners explore: how can the same radiation that damages DNA in a rock sample be used to save lives? This supports understanding of radiation properties and applications. – The Half-Life of Iodine-131 in Thyroid Treatment: Doctors prescribe a specific dose of radioactive Iodine-131 knowing it will reduce to safe levels within weeks. Learners explore: how can doctors predict exactly when a substance will become safe? This supports understanding of half-life and decay curves. – Kenya's Nuclear Energy Debate (KNEB): Kenya's government through the Kenya Nuclear Electricity Board has explored adding nuclear power to the national grid. Learners explore: how does splitting a uranium nucleus release enough energy to power Nairobi? This supports understanding of nuclear fission, energy release, and the societal dimensions of nuclear science.

LESSON 1 (80 minutes): The Invisible Killer in the Rock: Nuclear Structure, Isotopes & Launching the Phenomenon	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To confront learners with the anchoring phenomenon — pitchblende ore exposing photographic film in complete darkness — and to build the foundational nuclear vocabulary and conceptual structures (nucleon, nuclide, isotope, N vs Z plot, belt of stability) needed to begin explaining what is happening inside that rock.
Knowledge	Learners will know: (a) the structure of the atomic nucleus in terms of protons, neutrons, and nucleons; (b) the meaning of atomic number (Z), mass number (A), and neutron number (N); (c) the definitions of nuclide, radionuclide, and isotope; (d) that isotopes of the same element are chemically identical but may differ in nuclear stability; (e) the concept of the belt (valley) of stability and the role of the neutron-to-proton (N:Z) ratio in determining nuclear stability; (f) that radioactivity is the spontaneous emission of radiation from an unstable nucleus.
Skills	Learners will be able to: (a) write and interpret standard nuclide notation (AZX); (b) calculate N, Z, and A from nuclide data and plot points on an N vs Z graph; (c) distinguish stable from unstable nuclides using the belt of stability; (d) formulate investigable questions from an observed phenomenon; (e) collaboratively construct and annotate a Driving Question Board (DQB).
Attitudes	Learners will appreciate: (a) that science begins with careful observation and honest puzzlement; (b) that invisible phenomena can have real and measurable effects on people and the environment; (c) the value of asking precise questions as the foundation of scientific inquiry.
Key Inquiry Question	What makes some atomic nuclei stable while others are unstable and undergo radioactive decay?
Purpose in Storyline	This is the launch lesson. Learners encounter the anchoring phenomenon (pitchblende exposing film in darkness) for the first time, record their initial predictions, and discover that matter — specifically the atomic nucleus — holds a story they cannot yet fully explain. Every subsequent lesson in this sub-strand will return to this moment and add a new layer of explanation. By the end of Lesson 1, learners have the vocabulary and the N vs Z framework to begin asking the right questions about nuclear stability, and the DQB is open and populated with their genuine wonderings.
Safety Notes	No radioactive materials are handled by learners in this lesson. The pitchblende demonstration (or video/image substitute) must be conducted or sourced by the teacher only. Remind learners that actual radioactive sources are never to be touched or brought to school. Images and video footage of nuclear materials carry no radiation risk.

B. LESSON OVERVIEW

Lesson 1 launches the entire sub-strand by anchoring every concept in a single compelling mystery: a piece of pitchblende (uranium ore) has exposed a photographic film locked in a lightproof box — with no light, no heat, and no sound involved. Learners first predict what could cause this, drawing only on prior knowledge. They then build the nuclear vocabulary required to begin explaining the phenomenon (nucleon, nuclide, isotope, atomic number, mass number, neutron number, radioactivity). A structured investigation of Carbon-12 vs Carbon-14 — chemically identical twins with different nuclear fates — makes the isotope concept concrete. Small groups plot N vs Z graphs for a curated set of Kenyan-context nuclides and shade the belt of stability, discovering that nuclides outside the belt are the unstable ones responsible for the phenomenon. The lesson closes with the opening of the class Driving Question Board (DQB), where learners post their most burning questions. Kenya-relevant contexts (Mrima Hill monazite deposits, nuclear medicine at Kenyatta National Hospital, energy security debates) are woven throughout to ensure the phenomenon feels locally urgent and not merely academic.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Chronometric revolution Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Radioactivity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners enter to find a large photograph on the board: a photographic film showing a dark, rock-shaped silhouette — the classic Becquerel pitchblende image, contextualised as 'found in a geology lab in Nairobi'. On each desk is a sealed envelope labelled 'DO NOT OPEN'. The teacher narrates the scenario: the film was locked in a lightproof box next to a sample of Kenyan pitchblende ore; when developed, it showed this image. Learners individually write a 3-sentence prediction in their notebooks answering: (1) What do you think came out of the rock? (2) Where did that 'something' get its energy? (3) Why do you think only some rocks might do this? They then share predictions with a shoulder partner (Think-Pair-Share) before 4–5 cold-calls bring ideas to the whole class. Predictions are written verbatim on a 'Before' column of a class anchor chart — they will not be erased; they will be revisited every lesson.	Teacher narrates slowly and theatrically: 'This film was locked — no light could get in. No heat lamp, no torch, no phone screen. Just a rock. And yet… the film was exposed. Something came out of that rock.' [Pause 10 seconds — allow silence to build curiosity.] Teacher then says: 'In your notebook, I want THREE sentences. Sentence one: what do you think came out of the rock? Sentence two: where did that something get its energy? Sentence three: why might only SOME rocks do this?' [Allow 4 minutes of silent individual writing.] Teacher circulates, noting 3–4 interesting or misconception-revealing predictions to cold-call. After pair sharing (90 seconds), teacher cold-calls exactly 5 learners by name — aiming for gender balance and geographic diversity of seating. Teacher scribes each prediction word-for-word on the anchor chart without correcting. Teacher closes with: 'None of us needs to be right yet. Science starts exactly here — with a good puzzle and an honest guess. We will come back to YOUR predictions at the end of every single lesson this term.'	Think-Pair-Share with Anchor Chart Documentation. This strategy externalises prior knowledge and makes misconceptions visible and discussable. By committing predictions to paper before instruction, learners create a cognitive 'stake in the ground' that makes subsequent revision meaningful. The anchor chart becomes a living artefact of the class's growing understanding across all 7 lessons.	Teacher scans written predictions during circulation. Observable indicator of readiness: learner predictions reference at least one of — energy, particles, waves, atoms, or the rock's composition. A learner who writes only 'I don't know' needs immediate quiet prompting: 'Think about what we know makes things glow or give off energy — light, heat, sound… could there be something else?' Teacher notes which learners conflate radiation with heat or light — these misconceptions are flagged for targeted questioning in Phase 3.
Observe Phase 📹 **VIDEO:** Chronometric revolution Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners open the sealed envelope on their desks to find a one-page structured reading: 'Inside the Rock — A Story of Nuclei' (teacher-authored, Kenya-contextualised). The reading covers: atomic structure review (protons, neutrons, electrons); definitions of nucleon, nuclide,	Teacher says: 'Before we can explain the rock, we need a shared language. Open your envelope — you have 8 minutes to read and complete the table. Work silently first, then check with your group.' [8 minutes silent reading and table completion.] Teacher circulates to 4–5 groups, asking: 'Show me how	Data Table Completion + N vs Z Graph Construction (NGSS Science and Engineering Practice 4: Analysing and Interpreting Data; Practice 5: Using Mathematics and Computational Thinking). Learners construct the visual pattern (belt of stability) from raw data rather than receiving it as a diagram. This	Teacher checks completed N vs Z tables for two key indicators: (1) correct calculation of N = A − Z for all 10 nuclides (target: ≥ 8/10 correct before moving on); (2) correct identification that stable nuclides cluster in a band and that the band curves above the N = Z line for heavier elements. Groups

📖 **HTML:** [Radioactivity (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	atomic number Z, mass number A, neutron number N; standard nuclide notation AZX with worked examples (Carbon-12, Carbon-14, Uranium-238, Potassium-40 — the last two found in Kenyan geology). A second section introduces the concept of isotopes using the analogy of two Kenyan runners from Kericho — same name, same village, same training diet (ugali and sukuma wiki), but one carries an extra weight in their backpack. They look identical on the registration form, but one tires before the finish line. A data table lists 10 nuclides (including isotopes of Carbon, Potassium, Uranium, and Radium) with Z and A values. Learners calculate N for each, complete the table, and then plot each nuclide as a dot on a pre-printed N vs Z graph (N on y-axis, Z on x-axis). A reference line N = Z is pre-drawn. Groups shade the region where stable nuclides cluster — the belt of stability — and circle the outsiders. A brief embedded video clip (2 minutes, teacher-curated or described if offline) shows Marie Curie's pitchblende experiments alongside footage of Kenyan geologists at Mrima Hill in Kwale County, where monazite (a radioactive mineral) is found.	you calculated N for Uranium-238' and 'What do you notice about the N:Z ratio as Z gets larger?' [Allow 10 seconds wait time after each question before accepting responses.] After table completion, teacher models plotting the first two nuclides on the projected N vs Z graph, narrating: 'Carbon-12: Z = 6, N = 6. I plot it here, right on the N = Z line. Carbon-14: Z = 6, N = 8. I plot it here — above the line. Now you plot the remaining 8.' [4 minutes group plotting.] Teacher then asks the whole class: 'Groups, look at your graphs. Where do most stable nuclides sit? Where do the unstable ones sit?' Cold-calls 3 groups to share observations. Teacher annotates the projected graph as groups report, drawing and labelling the belt of stability. Teacher then says: 'Now look back at the rock in the photograph. Uranium-238 is outside the belt of stability. What does your graph suggest is happening inside that rock?' [Wait 12 seconds.]	transforms the belt of stability from an abstract textbook concept into a pattern they discovered themselves — directly linking to the phenomenon by placing Uranium-238 visibly outside the belt.	that have Uranium-238 plotted inside the belt of stability have made an error and need immediate redirection. Exit check: teacher asks one member from each group to point to where their group plotted Carbon-14 and explain in one sentence why it is or is not in the belt of stability.
Explain Phase 📹 **VIDEO:** [Chronometric revolution] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:**	Learners engage in a Jigsaw Expert Group activity. Four expert topics are assigned (one per group of 4): (A) What is radioactivity? — definition, spontaneous vs induced, who first described it (Marie Curie, Henri Becquerel); (B) Why does nuclear instability occur? — the role of the strong nuclear force vs electrostatic repulsion among protons, and why	Teacher introduces the Jigsaw: 'Each group becomes an expert in one piece of the puzzle. You have 7 minutes in your expert groups. Your job is to understand your topic well enough to teach it.' [7 minutes expert group work — teacher circulates and listens, intervening only to ask clarifying questions, not to give answers: 'Can you say that in a way your	Jigsaw Cooperative Learning followed by Synthesis Mini-Whiteboard Check (NGSS Practice 6: Constructing Explanations; Practice 8: Obtaining, Evaluating, and Communicating Information). The Jigsaw ensures every learner is both teacher and student, deepening retention. The mandatory use of 5 specific	Mini-whiteboard simultaneous reveal allows teacher to scan all groups in under 30 seconds. Success criteria: (1) the word 'nucleus' or 'nuclear' is used (not 'atom' alone); (2) 'neutron-to-proton ratio' or 'N:Z' appears; (3) 'belt of stability' is referenced; (4) 'unstable' is applied to the uranium nucleus specifically; (5) 'radioactivity' is described as

Radioactivity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	extra neutrons help stabilise up to a point; (C) Isotopes in Kenya's context — Potassium-40 in Kenyan soil and food (ugali, beans), Carbon-14 in dating ancient Kenyan artefacts found at sites like Olorgesailie; (D) The nuclide notation system — practice writing and reading AZX notation for 6 nuclides, converting between notation and descriptive language. Expert groups read, discuss, and prepare a 90-second verbal summary. Learners then regroup into mixed groups (one expert from each topic per group) and teach each other their expert content. Each mixed group then answers a synthesis question on a mini-whiteboard: 'Using what all four experts have shared, write two sentences explaining what is happening inside the pitchblende rock in the photograph — use the words: nucleus, neutron-to-proton ratio, belt of stability, unstable, and radioactivity.' Mini-whiteboards are held up simultaneously for teacher scan.	classmate who missed yesterday would understand?' and 'How does your topic connect back to the rock?'] Teacher then calls time and directs regrouping: 'Now move to your mixed groups — seats A, B, C, D. You have 8 minutes. Every expert teaches their piece. Then, as a group, answer the synthesis question on your mini-whiteboard.' [8 minutes mixed group work.] After whiteboards are raised, teacher selects 2 boards that correctly use all 5 required terms and reads them aloud, then selects 1 board with a partial answer and asks: 'This group is almost there — what is the one word or idea missing?' [Cold-call to a different group to complete it.] Teacher then delivers a 5-minute direct instruction consolidation: 'Let me sharpen what you've discovered…' — clarifying that radioactivity is spontaneous (the rock needs no outside energy input), that it is a nuclear — not a chemical — process, and that this is why it cannot be stopped by heating, cooling, or reacting the rock with acids.	vocabulary terms in the synthesis question forces learners to operationalise definitions in the context of the phenomenon rather than reproduce them in isolation.	spontaneous. Groups missing 2 or more criteria are seated next to a stronger group for the DQB phase and receive a targeted follow-up question at the start of Lesson 2.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Chronometric revolution Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Radioactivity (Flexbook) Source: CK-12	Each learner receives three sticky notes (or small paper squares if sticky notes are unavailable). They write one question per note — questions they genuinely cannot yet answer about the phenomenon or about radioactivity in Kenya. Sentence starters are offered on the board: 'I wonder why…', 'I want to know how…', 'I am confused about…', 'I wonder if Kenya…'. Learners post their questions on the class DQB (a large sheet of paper or section of the classroom wall divided into columns: Nuclear Structure / Types of Radiation /	Teacher says: 'You have 3 minutes. Three sticky notes. Three genuine questions — things you actually want to know, not things you think I want to hear. There are no wrong questions here. If you are wondering it, write it.' [3 minutes silent individual writing — teacher writes their own question on a sticky note to model that even the teacher has questions.] Teacher then says: 'Come up and place your questions in the column where you think they belong. If you're not sure which column, put it in the middle — we'll sort it	Driving Question Board Construction (NGSS Practice 1: Asking Questions and Defining Problems). The DQB makes the class's collective curiosity visible and persistent. By categorising questions into sub-strand topics before instruction, learners create their own advance organiser. The voting and 'Question of the Week' rituals build ownership and accountability — learners return to the DQB each lesson as scientists tracking what they now know vs what remains unresolved.	Teacher reads all posted questions before the next lesson. Observable indicators: (1) at least 70% of questions are genuinely investigable (not Google-answerable in one sentence) — if fewer, teacher adds 2–3 model questions to the board; (2) questions span at least 3 of the 7 DQB columns — if they cluster only in one area, teacher notes this as a gap in learners' mental model of the sub-strand's scope; (3) absence of questions from specific learners is noted — teacher follows up quietly at the

🔍 Search ARES for similar readings	Detection / Half-Life / Nuclear Equations / Applications in Kenya / Dangers and Protection). The teacher reads 6–8 questions aloud, grouping them into columns. The class votes (show of hands) on the top 3 'most burning' questions that they want to be sure this sub-strand answers. These three are starred and placed at the top of the DQB. One question from a learner that is particularly insightful or surprising is designated the 'Question of the Week' and displayed prominently.	together.' [2 minutes posting.] Teacher reads questions aloud, grouping and occasionally asking the question's author: 'Can you say a little more about what you mean here?' Teacher narrates: 'Look at this board. These are YOUR questions. Every lesson we come in, we will ask: what new evidence have we gathered to answer these questions? By the end of this sub-strand, we are going to be able to answer every single one.' Teacher designates the 'Question of the Week' — ideally one that connects to Kenya (e.g., 'Does the soil in Kwale near Mrima Hill make people sick?') — and says: 'This question is going to stay with us. Keep thinking about it.'		start of Lesson 2.
Model Building Phase 📹 **VIDEO:** Chronometric revolution Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Radioactivity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners individually draw two nuclear models side-by-side in their notebooks: the nucleus of Carbon-12 and the nucleus of Carbon-14 (using the standard proton-neutron sphere model). They annotate each diagram with: Z, N, A, the nuclide notation AZX, and a stability label ('STABLE — inside belt' or 'UNSTABLE — outside belt'). Below the diagrams, they write a Model Statement: 'The pitchblende rock emits invisible radiation because it contains _____ nuclei. These nuclei are unstable because their N:Z ratio is _____, which places them _____ the belt of stability. To become more stable, these nuclei _____.' The blanks are intentionally left incomplete for terms not yet taught (types of emission) — learners write 'TBC' for what they do not yet know. Groups share their completed model statements with an elbow partner and note any differences. The teacher collects	Teacher says: 'We are going to build a model of what is happening inside the pitchblende rock. A model is our best current explanation — it will be WRONG in some ways, and that is fine. Scientists revise their models. So will we.' Teacher displays a partially completed example model on the board (Carbon-12 only, with labels but no stability annotation). Teacher says: 'You complete Carbon-12 and then draw Carbon-14 yourself. For the Model Statement below, fill in what you know. For what we haven't learned yet, write TBC — To Be Confirmed.' [7 minutes individual drawing and writing.] Teacher circulates and asks 3 targeted questions to individual learners: 'Why does Carbon-14 have 8 neutrons and not 6?', 'If I told you the N:Z ratio for Carbon-14 is 1.33 — is that inside or outside the belt of stability for light elements?', and 'What do you predict a nucleus	Progressive Model Building with Intentional Incompleteness (NGSS Practice 2: Developing and Using Models). By deliberately leaving blanks labelled 'TBC', the model communicates to learners that scientific models are provisional and that incompleteness is intellectually honest — not a failure. The collected baseline model becomes a tangible record of conceptual growth across the sub-strand, making learning visible to both learner and teacher.	Teacher reviews collected/photographed model drawings before Lesson 2. Success criteria for Lesson 1 baseline model: (1) Carbon-12 and Carbon-14 are drawn with correct proton and neutron counts (6p+6n and 6p+8n respectively); (2) correct nuclide notation is written for both; (3) Carbon-12 is labelled STABLE and Carbon-14 is labelled UNSTABLE; (4) the Model Statement correctly identifies the N:Z ratio discrepancy and the belt of stability as the reference frame. Learners who have drawn the electron shells (not the nucleus) have a critical misconception — radioactivity is nuclear, not electronic — and must be addressed at the very start of Lesson 2 with a direct, brief correction.

	or photographs one model per group as a baseline artefact — these will be directly revised in Lessons 2, 3, and 4 as new content is learned.	does when it is outside the belt?' [Wait 12 seconds each.] Teacher closes by saying: 'Every lesson, we will update this model. By Lesson 7, you will be able to fill in every blank — and you will have built that understanding yourselves, from the evidence.'		

D. TEACHER REFLECTION

After Lesson 1, reflect on the following questions before planning Lesson 2:

1. PHENOMENON HOOK: Did the pitchblende photograph generate genuine puzzlement, or did learners seem indifferent? If indifferent, consider bringing a physical object (a chunk of dark rock, even non-radioactive granite) to make the phenomenon more tangible in Lesson 2.

2. VOCABULARY LOAD: Were learners overwhelmed by the number of new terms (nucleon, nuclide, isotope, nuclide notation, belt of stability, radioactivity) introduced in one lesson? If so, consider creating a class 'Nuclear Glossary Wall' where each term is added with a hand-drawn icon as it is introduced across the sub-strand.

3. DQB QUALITY: Read every question on the DQB carefully. Are learners asking questions about the nuclear level, or are they still asking surface questions about the rock itself? If the latter, open Lesson 2 by modelling a 'deeper' version of a surface question: 'A classmate asked "why is the rock black?" — a great observation question. A nuclear scientist would then ask "what is inside the atoms of that black rock that behaves differently from the atoms in white limestone?" How can we push our questions one level deeper?'

4. GENDER AND PARTICIPATION: Were the 5 cold-call slots in Phase 1 genuinely gender-balanced? Were girls as likely as boys to post questions on the DQB? If not, introduce a structured turn-taking protocol (e.g., alternating rows for cold-calls) starting in Lesson 2.

5. N vs Z GRAPH ACCURACY: How many groups correctly placed Uranium-238 outside the belt of stability? This is the single most important graphical outcome of Lesson 1 — it is the visual 'proof' that connects the phenomenon to the concept. Any group that placed it inside the belt needs a 5-minute reteach at the start of Lesson 2 before the class moves on.

6. KENYA CONTEXT RESONANCE: Did the references to Mrima Hill, Kenyatta National Hospital, and Kenyan geology generate interest or questions? Note which Kenya contexts generated the most energy — return to those in later lessons.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	A photographic film was exposed inside a sealed, lightproof box placed next to a sample of pitchblende (uranium ore) — with no light, heat, or sound present. The film showed a clear dark silhouette of the rock.
What did I learn?	Atoms contain a nucleus made of protons and neutrons (nucleons). The atomic number (Z), mass number (A), and neutron number (N = A − Z) define a nuclide, written as AZX. Isotopes are atoms of the same element with the same Z but different N — chemically identical but potentially very different in nuclear stability. Stable nuclides cluster in the 'belt of stability' on an N vs Z graph; nuclides outside this belt (like Uranium-238 and Carbon-14) are unstable. Radioactivity is the spontaneous emission of energy or particles from an unstable nucleus — it requires no external energy input and cannot be stopped by ordinary chemical means.
How does this explain the phenomenon?	The pitchblende rock exposes the film because it contains uranium nuclei (e.g., Uranium-238) whose N:Z ratio places them far outside the belt of stability. These unstable nuclei spontaneously release invisible energy and/or particles — radioactive emissions — as they attempt to reorganise into a more stable configuration. This is a nuclear process, not a chemical one, which is why the rock continues emitting radiation indefinitely without any external energy source. The type of emission and the rate of decay (half-

	life) are concepts we will investigate in the lessons that follow.

LESSON 2 (80 minutes): Types of Radiation: Alpha, Beta, and Gamma — Penetrating Power and Nuclear Identity	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate and compare the properties of alpha, beta, and gamma radiation — including penetrating power, charge, mass, speed, and ionising ability — and to connect each radiation type to the nuclear changes that drive a nucleus toward stability.
Knowledge	Learners will know that: (1) radioactive nuclei emit three types of radiation — alpha (α), beta (β), and gamma (γ); (2) each type differs in charge, mass, speed, ionising ability, and penetrating power; (3) alpha emission decreases both mass number (by 4) and atomic number (by 2); (4) beta emission increases atomic number by 1 without changing mass number; (5) gamma emission releases energy without changing nuclear composition; (6) these emissions shift unstable nuclei on the N vs Z plot toward the belt of stability.
Skills	Learners will be able to: (a) predict and compare the penetrating power of α, β, and γ radiation through paper, aluminium, and lead analogues; (b) tabulate the properties of the three radiation types; (c) write and interpret nuclear decay equations for α and β emissions; (d) represent α and β decay as directional arrows on an N vs Z stability diagram; (e) use evidence to construct and revise a cumulative nuclear stability model.
Attitudes	Learners will appreciate that: (a) the invisible radiation from pitchblende ore is not random chaos but a purposeful nuclear reorganisation governed by physical laws; (b) understanding radiation types is essential for protecting communities living near mineral-rich areas in Kenya such as the Rift Valley and Coast regions; (c) scientific curiosity and careful evidence gathering — as demonstrated by Marie Curie and Kenyan physicists — are central to advancing knowledge.
Key Inquiry Question	What makes some atomic nuclei stable while others are unstable and undergo radioactive decay, and how does each type of radiation emitted reflect the specific imbalance inside the nucleus?
Purpose in Storyline	In Lesson 1, learners established that the pitchblende rock contains uranium atoms with neutron-to-proton ratios outside the belt of stability, and that those nuclei spontaneously release invisible energy. But the film exposure from Lesson 1 tells us nothing about *what kind* of invisible emission is hitting the film. In Lesson 2, learners crack open that question: the rock is not emitting one thing — it is potentially emitting three distinct types of radiation, each with a different nuclear origin, a different ability to pass through matter, and a different hazard profile. By the end of this lesson, learners can point to the pitchblende and say: 'The uranium nucleus is too heavy — it sheds an alpha particle to lose mass and move toward stability; the resulting daughter nucleus may still be unstable and emit a beta particle to fix its neutron-to-proton ratio; and gamma rays carry away leftover energy.' This mechanistic picture drives every subsequent lesson in the sequence.
Safety Notes	No live radioactive sources are used in this lesson. All activities use diagrams, simulations, and physical analogues (sheets of paper, cardboard, and aluminium foil). Remind learners that in real laboratory or field settings — such as geological surveys in Kenya's Rift Valley — appropriate shielding, distance, and dosimetry are mandatory. Emphasise that gamma radiation requires lead or thick concrete shielding. Display the KEBS/KIRDI radiation safety poster if available.

B. LESSON OVERVIEW

Learners revisit the anchoring phenomenon — the pitchblende rock exposing photographic film — and are challenged to explain *what* is coming out of the rock. Through a structured Predict-Observe-Explain cycle, they use diagrams and a PhET-style simulation analogue to investigate how alpha, beta, and gamma radiations differ in their ability to penetrate paper, aluminium foil, and lead sheets. Working in groups, learners build a comparative properties table, write nuclear decay equations, and then return to their N vs Z stability model from Lesson 1 to draw decay arrows that show how α and β emissions physically move a nucleus across the plot toward

the belt of stability. The lesson closes with a Driving Question Board (DQB) update and a cumulative model revision that directly advances the class's ability to explain the anchoring phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** More on Normal force (shoe on floor) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Raising a Product or Quotient to a Power (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a projected image of the original pitchblende-and-film setup from Lesson 1. Alongside the image is a new diagram showing three barriers placed between the rock and the film: a sheet of paper, a sheet of aluminium foil, and a thick lead block. Each group receives a 'Prediction Card' with three columns (Paper / Aluminium / Lead) and three rows (Radiation Type A / B / C — types are not yet named). Without any instruction, learners individually predict: 'Which radiation type(s) will be stopped by each barrier? Tick the barrier that you think will block it.' After 3 minutes of silent individual prediction, groups of four share their reasoning and arrive at a consensus group prediction. Groups record their predictions on a class-shared whiteboard grid (physical or digital).	Display the pitchblende image and say: 'Yesterday we agreed that something invisible is leaving this rock and hitting the film. Today's question is: is it ONE thing, or could there be DIFFERENT types of invisible emission — each behaving differently?' Distribute Prediction Cards. Say: 'Work silently for 3 minutes. Do NOT look anything up. I want your honest best guess based on what you already know about matter and energy.' [WAIT — allow 3 minutes of silence.] After individual work, say: 'Now share with your group. You have 2 minutes to reach a consensus. Be ready to defend your prediction.' Cold-call 3 groups to share their consensus predictions, asking: 'What is your reasoning for why paper would or would not stop something coming out of a nucleus?' [WAIT 10 seconds after each response before probing.] Record all group predictions on the class whiteboard grid without correcting any errors — label them clearly as 'Our Predictions — Before Evidence'.	Prediction with Justification (Claim-Evidence-Reasoning primer): By committing to a prediction before instruction, learners activate prior knowledge about matter, energy, and penetration (e.g., light, sound, heat). Disagreements between groups create productive cognitive conflict that motivates the observation phase. Recording predictions publicly ensures accountability and makes conceptual change visible later.	Observable indicator: Can learners articulate a reason — even an incorrect one — for their penetration prediction? Listen for whether learners draw on particle size, energy, or charge as reasoning categories. Flag any group that simply guesses with no reasoning; seat the teacher near that group during the Observe phase. Collect Prediction Cards at the end of the phase to compare with post-lesson explanations.
Observe Phase 📹 **VIDEO:** More on Normal force (shoe on floor) Source: Khan Academy (English	Each group receives: (1) a printed 'Radiation Penetration Diagram Set' — three side-by-side diagrams showing a radiation source, then paper, aluminium, and lead barriers in sequence, with arrows indicating which radiation	Distribute materials and say: 'You have 4 minutes to study the diagrams and data card silently — no talking. I want you to notice three things: what is the charge of each radiation? How massive is it? How far does it travel through	Structured Data Table Construction: Learners extract information from multiple representations (diagram + data card) and organise it into a single comparative table. This builds the habit of evidence triangulation —	Observable indicator: Completed Evidence Tables should show correct charge (+2 for α, -1 for β, 0 for γ), correct penetration order (α < β < γ), and correct ionising ability order (α > β > γ). Circulate and check that learners have not

- US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Raising a Product or Quotient to a Power (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	types pass through each barrier (alpha stopped by paper, beta stopped by aluminium, gamma partially attenuated by lead); (2) a 'Properties of Radiation' data card giving the charge, approximate mass (in atomic mass units), relative speed, ionising ability, and penetrating power of α, β, and γ radiation; (3) a short 'Scientist Spotlight' card describing how Kenyan physicist Prof. Julius Mwangi Nzioka contributed to radiation safety research at the University of Nairobi's Department of Physics. Learners study the diagrams and data card silently for 4 minutes, then work in groups to complete an 'Evidence Table' with columns: Radiation Type | Symbol | Charge | Mass (amu) | Speed | Ionising Ability | Penetrating Power | Stopped By. They then compare their completed Evidence Table against their original Prediction Card and annotate differences in red pen.	matter?' [WAIT 4 minutes — circulate, do not answer questions, redirect with 'What does the diagram show you?'] After silent study, say: 'Now work as a group to complete your Evidence Table. Use only the materials I have given you — this is an evidence-gathering phase, not a lecture.' Circulate and cold-call individuals (not group spokespersons) using name cards: 'Amina, what charge did you record for an alpha particle and where in the diagram did you find that evidence?' [WAIT 12 seconds.] After tables are complete, say: 'Now compare your Evidence Table to your Prediction Card. Circle every cell where your prediction was wrong and write one word explaining why you think you were wrong.' Cold-call 2 learners to share a specific prediction error: 'Kamau, which prediction surprised you most and what does the evidence show instead?'	no single source is sufficient. Annotating prediction errors in red pen makes conceptual change a concrete, visible act rather than a passive correction, reinforcing metacognitive awareness of how scientific understanding is revised by evidence.	reversed ionising ability and penetrating power — this is the most common misconception. If more than 2 groups show this reversal, pause the class and ask: 'If something ionises strongly, does that mean it has more or less energy left to keep moving through matter?' [WAIT 10 seconds.] Use responses to address the misconception before Phase 3.
Explain Phase 📹 **VIDEO:** More on Normal force (shoe on floor) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Raising a Product or Quotient to a Power (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Part A — Nuclear Decay Equations (10 min): Each group receives a 'Nuclear Equation Card Set' with three worked examples of decay from uranium-238 and its daughters. Learners use the evidence table from Phase 2 to annotate each equation: 'Why does alpha decay reduce mass number by 4 and atomic number by 2?' and 'Why does beta decay increase atomic number by 1 without changing mass number?' They then attempt two unguided decay equations: (1) Radium-226 undergoes alpha decay — write the daughter nucleus. (2) Carbon-14 undergoes beta decay — write the daughter nucleus. Part B — Connecting Back to the Rock (10	Introduce Part A by saying: 'We now know the three types of radiation by their properties. Now we need to understand what is actually happening INSIDE the nucleus when each type is emitted — this is how we explain the pitchblende rock.' Distribute Nuclear Equation Card Sets. Say: 'Read the three worked examples. Then use your Evidence Table to annotate WHY the mass number and atomic number change the way they do. You have 5 minutes.' [WAIT — circulate and listen for explanations.] Cold-call one learner: 'Fatuma, when uranium emits an alpha particle, where does the alpha particle come from — is it created from nothing, or	Representation Mapping (Equation ↔ Diagram ↔ Phenomenon): Learners work across three levels of representation simultaneously — symbolic nuclear equations, graphical N vs Z plots, and the anchoring phenomenon of the pitchblende rock. Research shows that fluency in translating between representations is a core indicator of deep conceptual understanding in nuclear physics. Drawing decay arrows on the N vs Z plot makes the abstract equation tangible: learners literally see the nucleus 'moving' toward stability with each decay.	Observable indicator: (1) Decay equations — check that learners correctly conserve mass number and atomic number (atomic number of Radium-226 alpha decay daughter should be 86, mass number 222 — Radon-222; Carbon-14 beta decay daughter should be Nitrogen-14). (2) Decay arrows — alpha decay arrows must point diagonally toward the belt (−2N, −2Z); beta-minus decay arrows must point horizontally right (+1Z, −1N). Any arrow pointing away from the belt indicates a fundamental misunderstanding of the direction of nuclear stability — address immediately by asking: 'What is the purpose of the decay? Is the nucleus trying to become

	min): Groups return to their N vs Z stability diagram from Lesson 1. Using the decay equations just completed, they draw colour-coded arrows directly onto the N vs Z plot: a red arrow for each alpha decay (moving the nucleus -2 on the Z axis and -2 on the N axis, net diagonal toward stability) and a blue arrow for each beta decay (moving the nucleus +1 on Z, -1 on N — horizontal shift toward the belt). Learners write one sentence beneath the diagram: 'The pitchblende rock releases alpha and beta particles because ___________.'	was it already in the nucleus?' [WAIT 12 seconds — accept and build on the response.] After the unguided equations, cold-call two groups to write their answers on the board. For Part B, say: 'Now take out your N vs Z diagram from last lesson. You are going to show on that diagram what alpha and beta decay actually DO to the nucleus — how they move it.' Demonstrate one arrow on the projected class diagram, then say: 'Finish the remaining arrows for your decay chains and write that sentence about the pitchblende rock.' Circulate and prompt: 'Is your arrow pointing toward the belt of stability or away from it? What would it mean if the arrow pointed away?'		more or less stable?'
Driving Question Board (DQB) Creation 📹 **VIDEO:** [More on Normal force (shoe on floor)] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Raising a Product or Quotient to a Power (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner writes two sticky notes: (1) a GREEN note answering — 'What can I now explain about the pitchblende rock that I could not explain before today?' (minimum: mention one specific radiation type and one property); (2) an ORANGE note adding a new question that this lesson has raised for them (e.g., 'How do we actually detect these radiations if we can't see them?', 'If uranium keeps decaying, will it ever run out?', 'How do Kenyan hospitals use radiation safely?'). Sticky notes are placed on the class Driving Question Board under the columns 'Now I Can Explain' and 'New Questions'. The class does a 2-minute gallery walk to read peers' orange questions, and each learner silently stars the one question they are most curious about.	Say: 'We started this lesson not knowing what type of invisible emission was coming from the pitchblende rock. Take 3 minutes to write your two sticky notes — be specific. Don't just write 'alpha radiation' — write what you can NOW EXPLAIN about alpha radiation and the rock.' [WAIT 3 minutes.] Facilitate placement on the DQB. Then say: 'Do a silent gallery walk — read the orange questions and star the one that you most want answered.' After 2 minutes, tally the stars publicly and say: 'Our top question from today is ___________. Write it in your notebook. We will return to this as our sequence continues.' Preview: 'Lesson 3 will give us tools to actually DETECT these radiations — which will help us answer some of your starred questions.'	Driving Question Board — Metacognitive Progress Tracking: The DQB is not a display activity — it is a cumulative epistemic record of the class's evolving understanding. Green notes document knowledge consolidation; orange notes document productive uncertainty. The gallery walk and star-voting process democratises curiosity and ensures that student-generated questions (not only teacher-set objectives) shape the learning trajectory. This directly enacts the NGSS practice of 'Asking Questions and Defining Problems.'	Observable indicator: Green notes should reference specific properties (e.g., 'I can explain that alpha particles are stopped by paper because they are large, heavy, and highly ionising — they lose energy quickly'). Vague green notes (e.g., 'I learned about radiation') indicate surface-level processing — pull those learners aside at the start of Lesson 3 for a 2-minute retrieval check. Orange notes that connect to real Kenyan contexts (hospitals, mining, agriculture) indicate transfer thinking — celebrate these publicly.
Model	Each group retrieves their	Say: 'Your model from Lesson 1	Cumulative Model Revision	Observable indicator: The updated

Building Phase 📹 **VIDEO:** More on Normal force (shoe on floor) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Raising a Product or Quotient to a Power (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	cumulative Nuclear Stability Model poster from Lesson 1 (an N vs Z plot with the belt of stability drawn and uranium-238 plotted outside it). They now add four new layers to the model: (1) Colour-coded decay arrows for the uranium-238 → thorium-234 (alpha) → protactinium-234 (beta) decay chain, with each arrow labelled with the radiation type emitted; (2) A 'Radiation Properties Legend' box in the corner of the poster showing the symbol, charge, mass, and penetrating power of α, β, and γ; (3) Small shield icons — paper, aluminium foil, lead — placed next to each radiation type on the legend to show what stops each one; (4) One sentence connecting the model to the Kenyan context: 'In Kenya's Rift Valley, rocks containing uranium decay by emitting _______, which can be blocked by _______, meaning that workers and communities need to _________ to stay safe.' Groups display their updated models on the wall.	showed us WHERE uranium sits relative to the belt of stability. Today's evidence lets us show HOW it moves toward stability — the arrows are the story of the decay. Take 10 minutes to update your poster with all four additions. Every addition must be traceable to evidence from today's activity.' Circulate and prompt: 'Show me on the N vs Z plot exactly where thorium-234 lands after the alpha decay. Is it closer to the belt? How can you tell?' and 'What shielding would you recommend for a geologist doing field work near pitchblende in Kitui County?' [WAIT 10 seconds for responses.] In the final 3 minutes, do a rapid poster comparison — ask one group: 'What is the single most important thing your model now shows that it did not show at the start of Lesson 1?' Cold-call a second group to agree or disagree. Close by saying: 'By Lesson 7, this model will be complete enough to answer our full driving question. Today you added the types of radiation — you now know what is coming out of the rock and why.'	(Progressive Modelling): The poster is a living scientific model that grows across all 7 lessons, mirroring how scientific models are built iteratively through evidence. Each lesson's additions are visually distinct (different colours/layers), so learners can see their own knowledge growth. Requiring a Kenya-specific sentence on every update ensures that abstract nuclear physics is always grounded in local relevance — connecting to KICD's emphasis on Physics for real-life problem solving.	model poster should show: (a) correctly positioned daughter nuclei (Th-234 at Z=90, N=144; Pa-234 at Z=91, N=143) with arrows pointing toward the belt; (b) a legend with accurate properties for all three radiation types; (c) shield icons correctly matched to each radiation type; (d) a complete, specific Kenya-context sentence. Groups missing item (a) have a conceptual gap in decay equation mastery and need targeted support at the start of Lesson 3. Photograph each group's poster for the teacher's formative record.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners successfully distinguish between ionising ability and penetrating power — the most persistent misconception in this topic? If not, how will you address this in the opening 5 minutes of Lesson 3 before introducing detection methods? (2) Were learners' N vs Z decay arrows pointing consistently toward the belt of stability? If arrows were drawn incorrectly (pointing away from the belt), this suggests learners understand the mechanics of the equation but not the purpose of decay — consider using an analogy at the start of Lesson 3: 'Think of a boda-boda rider carrying an overloaded sack of maize. They throw off bags one by one until the load is balanced — the nucleus does the same thing with particles.' (3) Did the Kenya-context sentence in the Model Building phase produce substantive thinking about radiation protection, or did groups write generic responses? Collect the posters and read the sentences — if they are generic, introduce a structured sentence frame for Lesson 3: 'In [Kenyan place], people who work near [radioactive material] face [specific radiation type], which can be blocked by [shielding], so the recommended safety measure is [specific action].' (4) Count the orange DQB questions — did any learner ask about half-life, detection, or nuclear applications unprompted? These questions can be used as motivating hooks at the start of Lessons 3–6. (5) Note which groups' models are most complete and which are weakest — consider pairing strong and weaker groups during Lesson 3's collaborative detection activity to scaffold peer learning."

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed diagrams and data cards showing that alpha radiation is stopped by a sheet of paper, beta radiation is stopped by aluminium foil, and gamma radiation requires lead shielding. They observed that the radiation type with the greatest ionising ability (alpha) has the lowest penetrating power, and vice versa for gamma radiation. They also observed the uranium-238 → thorium-234 → protactinium-234 decay chain represented as arrows on an N vs Z stability plot.
What did I learn?	Learners learned that radioactive nuclei emit three types of radiation — alpha (α: charge +2, mass ~4 amu, low penetration, high ionisation), beta (β: charge −1, mass ~0 amu, moderate penetration, moderate ionisation), and gamma (γ: charge 0, mass 0, high penetration, low ionisation). Alpha decay reduces mass number by 4 and atomic number by 2, moving the nucleus diagonally toward the belt of stability on the N vs Z plot. Beta-minus decay increases atomic number by 1 and decreases neutron number by 1, shifting the nucleus horizontally toward the belt. Gamma decay releases energy without changing nuclear composition. These decay types explain what invisible emissions are leaving the pitchblende rock and why.
How does this explain the phenomenon?	Learners can now explain that the pitchblende rock continuously exposes photographic film because uranium-238 nuclei — which sit outside the belt of nuclear stability due to an excess of neutrons relative to protons — spontaneously emit alpha particles to reduce their mass and proton number, moving toward a more stable nuclear configuration. The resulting daughter nuclei may still be unstable and emit beta particles to further adjust their neutron-to-proton ratio. Gamma rays carry away excess nuclear energy during these transitions. Each emission type differs in its ability to penetrate matter: alpha particles are stopped by a sheet of paper, making them less externally hazardous but extremely dangerous if inhaled (as in mining dust from Kenyan mineral-rich regions); beta particles require aluminium shielding; and gamma rays require lead or thick concrete, posing the greatest external radiation hazard to Kenyan communities near uranium-bearing geological formations.

LESSON 3 (80 minutes): Detection of Radiation: Seeing the Invisible	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners investigate how invisible nuclear radiation from unstable atoms — like those in Kenyan pitchblende ore — can be detected, measured, and monitored using scientific instruments, and explain how these instruments underpin radiation safety and beneficial applications in Kenya.
Knowledge	Learners will be able to: (a) describe the working principle of a Geiger-Müller (GM) tube and counter; (b) explain how a cloud chamber makes radiation tracks visible; (c) define background radiation and identify its natural and artificial sources; (d) explain how detection technology is applied in Kenyan hospitals (e.g. KNH radiography) and environmental monitoring (e.g. KNRA).
Skills	Learners will be able to: (a) observe and interpret virtual GM counter readings and cloud chamber simulations; (b) distinguish background radiation counts from source-induced counts using data; (c) construct annotated diagrams of a GM tube and cloud chamber; (d) analyse radiation detection data to draw evidence-based conclusions about safety thresholds.
Attitudes	Learners appreciate that invisible phenomena can be rigorously measured; develop a responsible attitude toward radiation safety; and value the role of scientific instrumentation in protecting Kenyan communities and advancing healthcare.
Key Inquiry Question	How do scientists and doctors detect and measure radiation that is completely invisible to human senses — and why does that measurement matter for people living near Kenyan mining sites or receiving medical treatment at KNH?
Purpose in Storyline	In Lessons 1 and 2, learners discovered that some atomic nuclei (like uranium in pitchblende) are unstable and spontaneously eject α, β, and γ radiation. But a critical question remained on the DQB: 'If radiation is invisible, how do we even know it is there — or how much there is?' Lesson 3 answers that question directly. By exploring how GM tubes and cloud chambers convert invisible nuclear events into observable signals, learners gain the evidential tools needed to understand radiation dosimetry, safety limits, and the detection work carried out every day at Kenyan hospitals and the Kenya Nuclear Regulatory Authority. Every instrument principle connects back to the pitchblende phenomenon: the same radiation that exposed the photographic film in the dark is what triggers the GM tube click and leaves a white trail in the cloud chamber.
Safety Notes	No actual radioactive sources are handled in this lesson — all observations use virtual simulations and video. Teachers should nonetheless establish clear classroom norms: (1) Remind learners that real GM tubes used in field work require trained operators; reference KNRA licensing. (2) Discuss that background radiation is natural and omnipresent — learners should not be alarmed by background counts. (3) Emphasise that the photographic-film exposure in the anchoring phenomenon used sealed, shielded ore — reinforce that handling unknown rocks or minerals without protective equipment is unsafe. (4) If any learner has a medical history involving radiation therapy (e.g. at KNH or Aga Khan Hospital Nairobi), treat the topic with sensitivity and frame all discussion around safety protocols.

B. LESSON OVERVIEW

In this third lesson of the evidence-gathering sequence, learners move from knowing that radiation exists and what types are emitted (Lessons 1–2) to understanding how radiation is detected and quantified. Using a virtual Geiger-Müller counter simulation (PhET or curated YouTube demonstration) and a cloud chamber animation, learners observe how each instrument converts invisible nuclear events into measurable signals. They investigate background radiation — discovering that even without a source, the GM counter clicks — and connect this to naturally occurring radioactive materials in Kenyan geology (granite in the Rift Valley, uranium in Mrima Hill deposits). The lesson culminates in a structured discussion of real-world Kenyan applications: radiography at Kenyatta National Hospital (KNH), radiation monitoring by

the Kenya Nuclear Regulatory Authority (KNRA), and soil/water sampling in Kenyan agricultural research. Learners revise their Driving Question Board, moving answered questions to the 'Evidence' column and generating new questions about measurement uncertainty, dose limits, and safety — setting up Lessons 4 (half-life) and 5 (nuclear equations).

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Video: Energy Skate Park: Basics Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Detection of Radioactivity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a photograph of a Geiger-Müller counter (physical device) and a still image of a cloud chamber with white radiation tracks. Without any prior instruction, they individually write responses to two prompts in their science notebooks: (1) 'How do you think this device knows that radiation is passing through it — remember, radiation is invisible?' and (2) 'What do you think the white streaks in the cloud chamber photograph are made of — and why do they look different from each other?' After 3 minutes of individual writing, learners share predictions with an elbow partner (1 minute). The teacher then cold-calls 4 pairs — two for each prompt — to share predictions with the class. All predictions are recorded (without correction) on a shared 'Prediction Parking Lot' on the board. The teacher explicitly connects back: 'In our pitchblende rock, something invisible left a mark on photographic film. Today we are asking: what other ways can we catch that invisible something in the act?'	1. Display the two images side by side on the projector. Say verbatim: 'Look carefully at both images. Do not tell me what you already know — tell me what you genuinely think is happening. Write your own ideas first.' Allow 3 minutes of silent individual writing. 2. Say: 'Now share with your elbow partner. You each have 30 seconds.' 3. After sharing, cold-call exactly 4 pairs using a random name stick. For each pair, probe with: 'What in the image made you think that?' 4. Record all predictions on the board under 'Our Predictions' without affirming or correcting any. 5. Say: 'We will come back to these predictions at the end of class. Some of you will be surprised.' 6. Apply 10–15 seconds of WAIT TIME after each cold-call question before accepting answers. 7. Explicitly connect to the phenomenon: 'The pitchblende rock in our opening story exposed film in the dark. The same invisible radiation that did that is what these instruments are designed to catch. Hold that thought.'	Prediction Parking Lot — A visible, public record of pre-instruction ideas. This strategy activates prior knowledge, creates cognitive dissonance when predictions are later tested, and gives the teacher diagnostic data about misconceptions (e.g. 'the device must glow because radiation glows'). It also creates accountability — learners know they will revisit their predictions.	Observable indicator: All learners produce at least two written sentences per prompt before partner share. Teacher circulates during the 3-minute writing window and notes (without correcting): (a) learners who reference electricity/ionisation in the GM tube (building block present); (b) learners who conflate radiation tracks with light beams (misconception to address in Phase 2); (c) learners who leave prompts blank (flag for targeted questioning during Phase 2). Cold-call responses reveal the range of prior understanding across the class.
Observe Phase 📹 **VIDEO:** Video: Energy	PART A — Virtual GM Counter (25 minutes): Learners open the curated GM counter simulation on their ARES offline tablets (pre-	1. Before launching the simulation, say: 'Before we begin, remember our pitchblende rock. The radiation coming from that rock is what	Structured Observation Guide with Data Recording — Rather than open-ended watching, learners follow a scaffolded protocol that	Observable indicators: (a) Each group's observation guide shows three distinct count rates (background, source present,

Skate Park: Basics Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Detection of Radioactivity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	loaded PhET 'Radioactive Dating Game' or a curated YouTube offline video of a working GM tube). Working in groups of 3, they complete a structured observation guide: (i) BACKGROUND RADIATION: With no source present, they count GM tube clicks for 30 seconds (or note the count rate on the simulation). They record: 'Clicks with NO source = ___.' (ii) SOURCE ADDED: A virtual alpha source is brought near. They record the new count rate. Then a beta source. Then a gamma source. They record each. (iii) SHIELDING TEST: They insert a virtual sheet of paper, then aluminium foil, then lead between the source and the GM tube, noting how count rates change for each radiation type. Learners sketch and label a diagram of the GM tube cross-section (cylindrical tube, central electrode, argon gas filling, thin mica window, electrical circuit) as shown in the simulation or video. PART B — Cloud Chamber Simulation (10 minutes): Learners watch a 4-minute offline cloud chamber video (pre-loaded) showing alpha tracks (thick, straight), beta tracks (thin, curved), and gamma tracks (faint, scattered). They sketch the three track types and annotate: why is the alpha track thick? (high ionisation, heavy particle) Why is the beta track curved? (lighter, deflected by magnetic field if present). PART C — Kenya Connection	would trigger this GM tube. Your job is to be a detective — collect evidence, not just watch.' 2. Distribute the structured observation guide. Set a 25-minute timer visible to all groups. 3. Circulate every 4–5 minutes. At each group, ask one targeted question from this list (rotate questions across groups): — 'Why do you think the GM tube still clicks even when there is no source in the room?' — 'What does it tell you that paper blocks alpha but not gamma?' — 'If you were a radiographer at KNH, which radiation type would worry you most — and why?' 4. Apply 10–15 seconds WAIT TIME after each question. Cold-call at least 3 different groups during the cloud chamber segment to narrate what they see: 'Group near the window — describe the alpha track. Use shape, thickness, and direction.' 5. During the Kenya Connection reading, say: 'As you read, underline one fact that surprises you. You will share it in Phase 3.' 6. After all three parts, bring the class together and say: 'Let's return to the Prediction Parking Lot. Without me telling you the answers — which predictions are you now ready to revise, and why?'	mirrors real scientific measurement practice (control condition → variable introduction → shielding test). This strategy ensures learners collect comparable data across groups, experience the concept of background radiation empirically (rather than being told about it), and build the evidence base needed for Phase 3 explanation. The sketch-and-label task encodes the GM tube mechanism into long-term memory through dual coding (visual + verbal).	source + shielding) — groups with identical numbers across conditions have not engaged with the simulation and need redirection. (b) GM tube diagrams include at least 4 labelled components (window, gas, central electrode, circuit). (c) Cloud chamber sketches show visibly different track types for α, β, γ. (d) During circulating questions, learners can articulate WHY paper blocks alpha (not just that it does) — responses mentioning 'ionisation,' 'heavy,' or 'short range' indicate readiness to move to Phase 3.

	Reading (5 minutes): Learners read a 150-word teacher-prepared card: 'At Kenyatta National Hospital (KNH) in Nairobi, radiographers use GM-type detectors and dosimeters to monitor how much radiation staff and patients receive during X-ray and radiotherapy procedures. The Kenya Nuclear Regulatory Authority (KNRA) operates environmental monitoring stations in Nairobi, Mombasa, and near the Mrima Hill mineral deposits in Kwale County — measuring background radiation levels in soil, water, and air to protect Kenyan communities. Farmers at the Kenya Agricultural and Livestock Research Organisation (KALRO) in Kisumu use radiation-based techniques to study nutrient uptake in maize and beans.'			
Explain Phase 📹 **VIDEO:** Video: Energy Skate Park: Basics Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Detection of Radioactivity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	PART A — GM Tube Working Principle (class discussion, 10 minutes): Starting from their sketched diagrams, learners co-construct an explanation of how the GM tube works. The teacher facilitates a 'claim–evidence–reasoning' (CER) structure: learners write one sentence for each: Claim ('The GM tube detects radiation because…'), Evidence ('In our simulation, we observed…'), Reasoning ('This works because radiation causes…'). Groups share; the class converges on the key mechanism: ionising radiation enters the thin mica window → ionises the argon gas atoms → freed electrons accelerate toward the central anode under high voltage → avalanche of electrons produces a measurable electrical pulse → the counter registers one	1. Open Phase 3 by returning to the Prediction Parking Lot: 'Let's check our predictions. [Read one prediction aloud]. Now that you have evidence — who wants to revise this? Use your data to explain.' Cold-call 2 learners. Apply 15 seconds WAIT TIME before accepting. 2. For the CER activity, say: 'Do not just describe what you saw. Explain WHY it works. Your claim needs a mechanism, not just a description.' 3. During background radiation brainstorm, if learners are stuck, give a Kenya-specific prompt: 'Think about what Nairobi buildings are made of. Or think about what nutrients are in sukuma wiki. Could any of those contain naturally radioactive atoms?' 4. Cold-call 3 different learners	Claim–Evidence–Reasoning (CER) Framework — CER is a structured sensemaking strategy that prevents learners from stopping at description ('the counter clicked more') and pushes them toward mechanistic explanation ('the radiation ionised the gas, causing an electrical pulse'). By requiring three distinct components in writing before class discussion, CER ensures all learners engage in scientific reasoning — not just the most vocal students. The Kenya application synthesis extends CER into a real-world context, reinforcing that physics concepts are tools for solving Kenyan problems.	Observable indicators: (a) Written CER statements — teacher spot-checks 6 notebooks during the synthesis task; acceptable CER must include the word 'ionise' or 'ionisation' in the Reasoning component. (b) Background radiation brainstorm — groups should generate at least 2 distinct sources; groups naming only 'the rock' or 'the sun' (without specifics) need a follow-up probe. (c) KNH synthesis sentences — the phrase 'without detection devices' should lead to a consequence related to health or safety, not a trivial observation. (d) Exit check: teacher poses orally — 'In one word, what does a GM tube measure?' (Acceptable: radiation / ionisation / counts / clicks) before moving to Phase 4.

	'click.' PART B — Background Radiation Explanation (8 minutes): Learners revisit their 'clicks with no source' data. The teacher poses: 'Where is the radiation coming from if there is no source in the room?' Learners brainstorm in pairs (2 minutes), then share. The class identifies: cosmic rays from space, naturally occurring radioactive isotopes in building materials (concrete, granite — like the Rift Valley granite used in many Nairobi buildings), radon gas from soil, and trace radioactive isotopes in food (potassium-40 in sukuma wiki and githeri). PART C — Kenya Application Synthesis (7 minutes): Using the Kenya Connection card from Phase 2, learners individually complete a 3-sentence synthesis: 'At KNH, GM-type detectors are used to… This matters because… Without detection devices, we would not know…' Three learners read their syntheses aloud. Teacher connects explicitly back to the phenomenon: 'The pitchblende ore that exposed that photographic film in the dark — if that rock were sitting in a Kenyan home, a GM tube would tell us exactly how dangerous it is. Detection turns invisible risk into actionable numbers.'	(those not yet called on today) to share background radiation sources. 5. After KNH synthesis sharing, say: 'I want to push your thinking: if KNRA monitors radiation near Mrima Hill in Kwale, what are they trying to find out — and what would a dangerously high count rate tell them?' Apply 10 seconds WAIT TIME. Accept 2 responses. 6. Explicitly connect to the driving question: 'We now have evidence for the second part of our driving question — how can we protect ourselves? Detection is the first step. You cannot protect yourself from what you cannot measure.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Energy Skate Park:	Learners return to the class Driving Question Board, which has been visible since Lesson 1. Working in their groups, they receive three types of sticky notes: GREEN ('We can now answer this'), YELLOW ('We have partial	1. Reintroduce the DQB: 'Every lesson, we are building a more complete answer to our driving question. Today was about detection — seeing the invisible. Let's see how that changes our board.'	Colour-Coded DQB Annotation — Using three visually distinct sticky note colours (green/yellow/orange) transforms the DQB from a passive display into an active evidence-tracking tool. The colour system requires learners to	Observable indicators: (a) Each group places at least 1 green and 1 orange sticky note — groups with no green notes may not have connected Phase 3 explanations to DQB questions; intervene with a prompt: 'Look at the question

Basics Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Detection of Radioactivity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	evidence'), and ORANGE ('New question this lesson raised'). Groups spend 6 minutes reviewing the DQB and placing their sticky notes. Expected GREEN moves: 'How do we know radiation is there if we can't see it?' (answered: GM tube, cloud chamber, photographic film). Expected YELLOW moves: 'How do we know how much radiation is safe?' (partial: we know detection gives counts, but we haven't yet linked counts to dose or safety thresholds — that comes in Lesson 6). Expected ORANGE additions: 'How many clicks per second is dangerous?', 'Does the rock ever stop emitting radiation?', 'Can radiation from Mrima Hill reach Nairobi?', 'If potassium-40 is in sukuma wiki and we eat it, does that hurt us?' The teacher facilitates a 4-minute whole-class gallery walk of the new additions. Two groups briefly narrate their most interesting ORANGE question and explain why it arose from today's evidence.	2. Distribute sticky notes and say: 'Be honest. Green only goes up if you can actually explain the answer using today's evidence — not just because we discussed it. Can you explain the mechanism? Then it's green.' 3. During the 6-minute group work, circulate and listen for productive ORANGE questions. Privately encourage specific learners: 'That question about sukuma wiki and potassium-40 is excellent — put it on the board. That is exactly what scientists at KALRO Kisumu are studying.' 4. During gallery walk, cold-call 2 groups to narrate their ORANGE question. Ask: 'What evidence from today made you wonder about that?' Apply 10 seconds WAIT TIME. 5. Close DQB phase by pointing to the driving question banner: 'We are getting closer. We know what radiation is (Lesson 1), what types exist (Lesson 2), and now how to detect it (Lesson 3). What do we still not know?' Pause 15 seconds. Accept 2–3 whole-class responses. 6. Say: 'Next lesson, we will tackle one of the orange questions that many of you wrote: does the rock ever stop? That will take us into half-life.'	metacognitively evaluate their own understanding (green vs. yellow) rather than defaulting to 'we learned this so we know it.' Orange notes are particularly powerful: they document productive confusion, model scientific curiosity, and directly seed the inquiry questions for subsequent lessons — making the storyline feel student-driven rather than teacher-imposed.	about seeing radiation — can you answer that now?' (b) Orange questions should be mechanistic or quantitative (e.g. 'How many counts is dangerous?') rather than purely factual (e.g. 'What is a GM tube?') — quality of orange questions indicates depth of engagement. (c) Narrating groups can justify their green classification with a causal explanation, not just a topic label.
Model Building Phase 📹 **VIDEO:** Video: Energy Skate Park: Basics Source: PhET Interactive Simulations (English)	Learners retrieve their cumulative nuclear model from Lesson 2 (a drawn cross-section of the pitchblende rock showing unstable uranium nuclei emitting α, β, and γ radiation with penetration arrows). They now add a Detection Layer to their model:	1. Distribute learners' Lesson 2 models (collected and stored). Say: 'Your model from last lesson showed radiation leaving the nucleus. Today we are adding the next layer: how does the outside world detect that radiation? Add the detection instruments to your existing model — do not start over.'	Cumulative Model Revision — Rather than building a new model each lesson, learners continually annotate and extend a single evolving diagram. This strategy makes conceptual progression visible and tangible: learners can literally see how their understanding has grown from 'the rock emits radiation' (Lesson 1) to	Observable indicators: (a) The GM tube diagram includes at minimum: mica window, argon gas, central electrode, counter display — 3 of 4 components is acceptable; fewer than 3 requires teacher support. (b) Penetration arrows correctly show α stopped before or at the GM tube window, β reaching the tube with reduced intensity, γ

🔍 Search ARES for similar videos 📖 **HTML:** Detection of Radioactivity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	(1) They draw a GM tube positioned outside the rock, with a labeled mica window facing the rock, argon gas inside, and a central electrode connected to a counter. They draw dotted arrows showing which radiation types (α, β, γ) reach the GM tube through different shielding materials — showing that α is stopped by the rock itself and the air gap, β reaches the tube but is attenuated, and γ passes through most barriers and is detected. (2) They add a small inset cloud chamber diagram showing the three track types (thick straight α, thin curved β, faint scattered γ). (3) They annotate the model with a 'Background Radiation Note' — a small text box explaining that even without the pitchblende, some counts occur from cosmic rays and natural radioactivity in the surrounding environment. (4) They write a two-sentence 'Model Limitation' note: 'My model does not yet show how to calculate how dangerous this radiation is, or how long the rock will keep emitting. We will add this in Lessons 4 and 5.' Learners share their revised model with their elbow partner and identify one difference between their two models — they explain whether that difference represents a mistake or a valid alternative representation.	2. Display an annotated reference diagram of a GM tube on the projector throughout the model-building time (8 minutes). Say: 'You do not need to memorise this — you need to show how it connects to the rock.' 3. Circulate and prompt with: 'Which radiation type from your uranium nucleus actually reaches the GM tube? Show that with your arrows.' Apply 10 seconds WAIT TIME after posing to any individual learner. 4. For the Background Radiation Note, say: 'Even in a Nairobi classroom, your model should show some radiation. Where does that come from? Show it.' 5. After partner sharing, cold-call 2 pairs to share one difference they found. Ask: 'Is that difference a mistake — or can both versions be correct?' Accept and discuss. 6. Collect all models to return in Lesson 4. Say: 'Next lesson, you will add the concept of half-life to this model — showing how the rock changes over time. Keep your model growing.'	'specific types of radiation travel different distances and are detected by specific instruments' (Lesson 3). The Model Limitation note is a critical element of scientific thinking — it teaches learners that a good model honestly acknowledges what it does not yet explain, rather than pretending to be complete.	passing through — reversal of α and γ penetration is a key misconception to flag and correct in Lesson 4 review. (c) Background radiation note names at least one specific source (cosmic rays, building materials, radon, potassium-40 in food). (d) Model Limitation note references a future lesson concept — this demonstrates that learners understand their model is incomplete and science is ongoing. (e) Partner comparison identifies at least one genuine structural difference, not just a cosmetic one (e.g. different arrow direction, different number of labelled components).

D. TEACHER REFLECTION
After delivering this lesson, reflect on the following questions in your professional journal: 1. BACKGROUND RADIATION SURPRISE: Did learners appear genuinely surprised that the GM counter clicks even with no source present? If not, the predict phase may need a more striking demonstration in future iterations — consider starting with the actual background count rate displayed on-screen before any discussion, letting the unexplained clicks provoke curiosity organically. 2. KENYA CONTEXT UPTAKE: During the KNH/KNRA reading and synthesis, did learners engage meaningfully with the Kenyan applications — or did they treat them as peripheral to the 'real' physics? If the latter, consider inviting a KNRA or KNH radiation safety officer as a virtual guest speaker in Lesson 6 (applications) to make the professional context vivid and credible. 3. DQB QUALITY: Review the orange sticky notes collected today. Are the new questions mechanistic and quantitative (e.g. 'How many counts per second is harmful?') or superficial (e.g. 'What does GM stand for?')? The quality of orange questions is your best diagnostic of whether learners are thinking like scientists or passive recipients. If questions are superficial, dedicate 5 minutes at the start of Lesson 4 to modelling what a 'deep' vs. 'surface' question looks like. 4. CER ASSESSMENT: When spot-checking notebooks during Phase 3, how many learners used the word 'ionise' or 'ionisation' correctly in their Reasoning component? If fewer than 60% did, plan a 5-minute ionisation review at the start of Lesson 4 using the analogy of a matatu knocking passengers off a footpath — the alpha particle (heavy, fast matatu) clears more passengers (electrons) from their atoms than a bicycle (beta particle). 5. MODEL PROGRESSION: Are learners treating the cumulative model as a meaningful artefact, or as a compliance task? Look for evidence of personalisation (added colour, additional annotations, self-generated labels). Learners who personalise their models are internalising the science. Learners who copy the reference diagram verbatim are not yet making it their own — consider assigning a 'model author statement' in Lesson 4: one sentence explaining what the model means to them personally.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed (via virtual simulation and video): (1) a Geiger-Müller counter producing clicks even with no radioactive source present (background radiation); (2) count rates increasing when alpha, beta, and gamma sources were brought near the GM tube, with different responses to shielding materials (paper, aluminium, lead); (3) cloud chamber tracks showing thick straight lines for alpha particles, thin curved lines for beta particles, and faint scattered trails for gamma radiation; (4) images and descriptions of GM-type detectors and dosimeters used at Kenyatta National Hospital (KNH) and environmental monitoring stations operated by the Kenya Nuclear Regulatory Authority (KNRA) near Mrima Hill, Kwale County.
What did I learn?	Learners learned that: (1) a Geiger-Müller tube detects radiation because ionising radiation enters through a thin mica window, ionises argon gas atoms inside the tube, and the freed electrons accelerate toward a central anode under high voltage, producing an electrical pulse registered as a 'click'; (2) background radiation is always present in the environment — sourced from cosmic rays, naturally radioactive materials in building stone (e.g. Rift Valley granite), radon gas from soil, and trace radioisotopes in food (e.g. potassium-40 in sukuma wiki and githeri); (3) a cloud chamber makes radiation tracks visible by allowing particles to ionise supersaturated alcohol vapour, forming condensation trails whose shape reveals the particle type; (4) detection instruments are essential tools for radiation safety in Kenyan hospitals, mining regions, and agricultural research, allowing invisible nuclear emissions to be converted into actionable, quantitative data.
How does this explain the	Learners explained that: (1) the invisible radiation from the pitchblende ore in the anchoring phenomenon — the same radiation

phenomenon?	that exposed photographic film in the dark — is detectable by a GM tube because it ionises matter (gas atoms in the tube) as it passes through, converting a nuclear event into an electrical signal; (2) different radiation types (α, β, γ) produce different GM count-rate responses with shielding because they differ in mass, charge, and ionising power — alpha particles are stopped by paper and air, beta particles by a few millimetres of aluminium, and gamma rays require dense lead shielding; (3) background radiation means that a GM tube will always record some counts even in the absence of a known source, so measurements of a radioactive sample must always be compared against the background count rate to determine the true emission from the source; (4) the rock in the anchoring phenomenon poses a quantifiable risk — detection devices give the numerical count rates that allow scientists, doctors, and regulators at KNRA and KNH to set safe exposure limits and protect Kenyan communities.

LESSON 4 (40 minutes): Half-Life and Decay Curves: How Long Before the Rock Goes Quiet?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners will quantify the rate at which unstable nuclei in the pitchblende rock decay over time, using the concept of half-life, so they can explain why the rock never seems to stop emitting radiation and how scientists use decay rates to date ancient Kenyan artefacts and calculate safe medical doses.
Knowledge	(1) Half-life is the time taken for half the radioactive nuclei in a sample to decay; (2) radioactive decay is random and spontaneous but follows a predictable statistical pattern described by a decay curve; (3) the equation N = N0 x (one-half)^n relates remaining nuclei N to initial nuclei N0 after n half-lives; (4) different isotopes have vastly different half-lives ranging from microseconds to billions of years; (5) Carbon-14 half-life of 5730 years is used to date organic remains at sites such as Olorgesailie; (6) Iodine-131 half-life of 8 days is exploited in thyroid treatment at Kenyatta National Hospital.
Skills	(1) Conduct and record a dice-simulation of radioactive decay; (2) plot a decay curve from experimental data; (3) extract half-life graphically from a decay curve; (4) apply N = N0 x (one-half)^n to solve quantitative decay problems; (5) connect decay-rate data back to the anchoring phenomenon and the N vs Z stability model.
Attitudes	Appreciate that mathematical models derived from simple probability experiments can describe and predict real nuclear behaviour; develop confidence in using quantitative tools to answer scientific questions about invisible processes.
Key Inquiry Question	What makes some atomic nuclei stable while others are unstable and undergo radioactive decay? (The half-life concept reveals that instability is not binary but a question of rate — some nuclei decay in an instant while others, like uranium-238 in the pitchblende rock, take billions of years, which is why the rock appears to emit radiation indefinitely on a human timescale.)
Purpose in Storyline	Lesson 4 moves learners from qualitative detection (Lesson 3) to quantitative prediction. Having confirmed that the pitchblende rock emits radiation and that instruments can detect it, learners now ask: at what rate does it decay, and how long will it keep doing so? The dice simulation makes the statistical nature of decay tangible; the decay curve ties raw data to the mathematical model; and the Carbon-14 and Iodine-131 contexts anchor the abstract equation in Kenyan reality. Decay curves are added to each group cumulative model, advancing the evidence base needed to answer the Driving Question fully in Lesson 7.
Safety Notes	No radioactive materials are used. Dice are safe for all learners. If a PhET simulation is used on devices, ensure screens are positioned so all group members can observe without crowding. Remind learners that the decay equation and half-life values they use today describe real isotopes whose radiation requires the shielding and distance rules discussed in Lesson 3.

B. LESSON OVERVIEW

Learners simulate radioactive decay by rolling 100 six-sided dice per group, removing every die that lands on 6 (representing a decayed nucleus) after each roll, recording surviving counts, plotting a decay curve, and extracting the half-life graphically. They then apply the equation N = N0 x (one-half)^n to solve two Kenya-contextualised problems: estimating the age of organic material at Olorgesailie archaeological site using Carbon-14, and calculating the remaining activity of Iodine-131 after a patient treatment cycle at KNH. Groups update their cumulative nuclear stability models with annotated decay curves and post new or answered questions to the Driving Question Board.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Worked example: Using oxidation numbers to identify oxidation and reduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Nuclear Decay Processes (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher displays a large photograph of pitchblende ore alongside a question projected on the board: 'If the rock in our phenomenon has been emitting radiation since before any human walked the Rift Valley — why has it not run out of radioactive atoms yet? Will it ever stop?' Learners spend 2 minutes writing individual silent predictions in their science journals. They then share in pairs (think-pair-share). The teacher collects three or four predictions verbally and records key phrases on the board without evaluating them: sample responses typically include 'maybe it never runs out', 'it slows down gradually', 'maybe new atoms form', or 'it depends on how many atoms there are'. The teacher then asks a bridging question: 'In Lesson 3 you measured background radiation — it was always there, always changing slightly. How might we describe the RATE at which atoms decay?' Learners write a second one-sentence prediction about whether decay rate stays constant, increases, or decreases over time.	Display the pitchblende photograph and the question simultaneously. Read the question aloud, then say: 'Take two full minutes — no talking, just write your own idea in your journal. I want YOUR thinking before we share.' START a visible 2-minute timer. After the timer: 'Now share with your partner for 90 seconds.' After sharing, cold-call three pairs using a random name stick. Write each contribution verbatim on the board under the heading PREDICTIONS — do not say correct or incorrect. Then say: 'Hold those predictions — we are going to gather evidence right now that will let you judge them yourselves.' WAIT TIME INSTRUCTION: After posting the bridging question about decay rate, wait a full 10 seconds of silence before accepting any response. If silence persists beyond 10 seconds say: 'Talk to your partner for 30 seconds, then we will hear from two groups.'	Elicitation of prior conceptions via think-pair-share; public commitment to predictions anchors motivation for the observation phase. Written predictions create a personal accountability record that learners will revisit in the Explain phase.	Teacher scans journals during the 2-minute silent write to check whether learners are engaging with the rate concept or only the quantity concept. Listen for misconceptions during sharing: (1) 'the rock makes new radioactive atoms' — flag this for resolution in the Explain phase; (2) 'the rate stays perfectly constant' — the simulation will challenge this at the individual-roll level while confirming the statistical trend. Note which pairs produce the most sophisticated predictions for targeted questioning later.
Observe Phase 📹 **VIDEO:** Worked example: Using oxidation numbers to identify oxidation and reduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Groups of four receive 100 six-sided dice each (or use an online dice-roller on one shared device if physical dice are unavailable). Each group also receives a printed data table with columns: Roll Number, Number of Dice Remaining, Number Removed This Roll. The protocol is explained: each die represents one radioactive nucleus; a result of 6 means the nucleus has decayed	Before starting, do a brief whole-class demonstration: shake 10 dice in a cup and spill them. Remove the 6s. Ask: 'Is this fair? Could a nucleus choose to decay?' After a 5-second wait say: 'Exactly — no choice. It is random. Each nucleus faces the same probability each moment. That is the key idea.' Circulate during the rolling activity. At round 3 pause the class and ask: 'Does the number	Hands-on analogue simulation makes the abstract concept of probabilistic decay concrete and observable. Graphing transforms raw counts into a visual pattern. Cross-group comparison of half-life values builds understanding of statistical consistency emerging from random events — a core insight of quantum-mechanical decay.	Circulate and check data tables after Round 5 for arithmetic errors — specifically learners who add removed dice back (a common procedural error). Check graphs for correct axis labels, appropriate scales, and smooth curves rather than dot-to-dot lines. Listen for whether learners articulate that the NUMBER removed decreases each round even though the PROBABILITY per nucleus stays

📖 **HTML:** [Nuclear Decay Processes (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	— remove that die; all other results mean the nucleus is still undecayed. Round 0 is recorded as 100 dice remaining. Groups roll all remaining dice simultaneously, count and remove every 6, record the survivors, and repeat for 10 rounds. One learner rolls, one counts survivors, one counts and removes decayed dice, one records — roles rotate each round. After completing the data table, groups use graph paper (or a shared spreadsheet if devices are available) to plot Roll Number (x-axis, 0 to 10) against Number of Dice Remaining (y-axis, 0 to 100). They draw a smooth best-fit curve through their points. They then draw a horizontal dashed line at exactly 50 dice remaining and drop a vertical line to the x-axis to read off the number of rolls it took for the sample to halve — this is their experimental half-life in roll-units. The teacher then connects roll-units to real time: 'If each roll represented 1000 years, what would the half-life be in years?' Groups record their extracted half-life and compare across groups, noting that all groups get approximately the same value (around 3.8 rolls, since each die has a 1-in-6 chance of decaying, giving a theoretical half-life of ln2 divided by ln(6/5) which is approximately 3.8 rolls) despite different individual data sets.	removed each round stay the same or change?' Wait 8 seconds. Listen for 'it decreases because there are fewer dice left' — affirm this without revealing the equation yet: 'Interesting. Why might that be? Talk to your group for 60 seconds.' After groups finish graphing, ask: 'Did every group get exactly the same decay curve?' After 5-second wait: 'No. Why not? What does the scatter tell us about individual nuclei?' Guide toward the idea that individual decay events are unpredictable but the population behaviour is predictable. Then introduce the vocabulary: 'The time for HALF the sample to decay is called the HALF-LIFE. What did you measure as your half-life in rolls?' Collect values from all groups and write them on the board — typically ranging from 3 to 5 rolls. Ask: 'Why the variation? What would happen if we used 10 000 dice instead of 100?' WAIT 10 seconds before taking responses.		constant — this distinction is the conceptual core of half-life. Use targeted questioning: 'If you started with 200 dice, would your half-life change?' (It should not — half-life is independent of initial sample size.)
Explain Phase 📹 **VIDEO:** [Worked example: Using oxidation numbers to identify oxidation and reduction] Source: Khan	The teacher introduces the formal equation N = N0 x (one-half)^n on the board, defining each symbol explicitly: N is the number of undecayed nuclei remaining, N0 is the initial number of undecayed nuclei, and n is the number of half-lives elapsed. Learners first verify	Write the equation slowly on the board, pointing to each symbol. Say: 'Before I explain this equation, use your dice data to GUESS what each symbol means — write your guesses in your journal. You have 90 seconds.' After 90 seconds, cold-call two	Equation-first-then-application with verification against own data creates cognitive coherence. Kenya-specific problems (Olorgesailie, KNH) situate abstract mathematics in authentic professional contexts. The journal paragraph forces explicit	Collect three or four journal paragraphs as exit-check proxies — look for whether learners correctly interpret 4.5 billion year half-life as meaning that only a tiny fraction of uranium-238 nuclei have decayed since Earth formed, keeping the rock active. Common

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Nuclear Decay Processes (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	the equation with their own dice data: using their recorded N0 = 100 and their experimentally extracted half-life, they calculate predicted N after 2 and 4 half-life periods and compare with their actual graph. The class then works through two Kenya-contextualised problems collaboratively. PROBLEM 1 (Carbon dating at Olorgesailie): Archaeologists at Olorgesailie prehistoric site in the Rift Valley find a charcoal fragment. They measure that the sample contains only one-eighth of the Carbon-14 activity of a living tree today. Carbon-14 has a half-life of 5730 years. How many half-lives have elapsed, and how old is the charcoal? Learners reason that one-eighth equals (one-half)^3, so n = 3, giving an age of 3 x 5730 = 17 190 years. PROBLEM 2 (Iodine-131 at KNH): A patient at Kenyatta National Hospital receives a dose of Iodine-131 for thyroid cancer treatment. The initial activity corresponds to 8000 undecayed nuclei (simplified unit). Iodine-131 has a half-life of 8 days. How many undecayed nuclei remain after 32 days? Learners calculate n = 32 divided by 8 = 4 half-lives; N = 8000 x (one-half)^4 = 8000 x (one-sixteenth) = 500 nuclei. Learners then write a connecting paragraph in their journals: 'Using what I know about half-life, I can now explain why the pitchblende rock in our phenomenon has been emitting radiation for so long without running out. Uranium-238 has a half-life of approximately 4.5 billion years, which means...' Pairs share their paragraphs before the teacher draws out the class	learners. Then confirm or redirect. For Problem 1, do NOT solve it for the class. Say: 'Work in your groups — you have 3 minutes. I will not help yet.' After 3 minutes say: 'Who got n = 3? How? Explain your reasoning to us.' WAIT 5 seconds for a volunteer before cold-calling. For Problem 2, say: 'This time one person in each group should try alone first, then compare with partners.' After 2 minutes: 'Did everyone agree? If you disagreed, explain where your reasoning diverged.' For the journal paragraph, say: 'Connect back to the ROCK. The uranium-238 in pitchblende has a half-life of 4.5 billion years. What does that number mean for why the rock seems to never stop emitting?' WAIT 12 seconds of silence — this is intentional productive struggle time. If no one responds say: 'Turn to a partner. You have 60 seconds to construct one sentence together.' After pairs share, say: 'Does this change any of the predictions you wrote at the start of today? Take 30 seconds and look back at what you wrote.'	connection back to the anchoring phenomenon, completing the predict-observe-explain reasoning arc.	error to probe: learners who write 'the uranium never decays' rather than 'the uranium decays very slowly so enormous numbers of nuclei remain undecayed'. During Problem 2, watch for the error of dividing N0 by 4 (number of half-lives) rather than applying (one-half)^4 — address this by asking: 'After one half-life, how many remain? After the second half-life, do you halve the original or the new number?'

	consensus explanation.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Worked example: Using oxidation numbers to identify oxidation and reduction] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Nuclear Decay Processes (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The teacher directs learner attention to the class Driving Question Board which has been accumulating since Lesson 1. Each group reviews the posted sticky notes silently for 90 seconds, then discusses: (1) Which questions can we now fully or partially answer using half-life evidence? (2) What NEW questions has today raised that were not on the board before? Each group writes at least one new question on a sticky note and one answered question response on a different coloured note. Groups post these to the board. The whole class does a 2-minute gallery walk to read new additions. The teacher then calls attention to one or two of the most generative new questions — likely questions such as: 'Can we use half-life to find out when the uranium in Kenyan rocks first formed?', 'Why does Carbon-14 keep forming in the atmosphere if it decays?', or 'Could a hospital waste Iodine-131 sample still be dangerous after 80 days?' These questions are explicitly labelled as evidence bridges to Lesson 5 (nuclear equations) and Lesson 6 (fission, fusion, and energy applications).	Say: 'Look at the questions we posted in Lesson 1 when we first saw the film exposed by the rock. Read them again now. Which ones could the ROCK answer today — using what you know about half-life?' Point physically to the board. After 90 seconds of silent reading: 'Groups discuss for 2 minutes. I want two things: one question you can now answer, and one brand-new question today raised for you.' Circulate and listen — prompt groups who are silent: 'Think about the Olorgesailie problem — what does it suggest you could do if you knew the half-life of ANY isotope in a rock?' After posting, during the gallery walk, stand near the board and use annotation: circle two new questions that will be answered in upcoming lessons and say: 'These two questions are going to drive the next two lessons. Keep them in mind.' WAIT TIME INSTRUCTION: When asking groups to share new questions, after one group shares, wait 6 seconds before inviting the next — this signals that thinking, not speed, is valued.	DQB update externalises metacognitive progress, making learning visible to the whole class. The practice of distinguishing answered from open questions builds scientific epistemic habits. Explicit bridging to future lessons sustains narrative coherence across the seven-lesson sequence.	Teacher photographs the DQB after updates. Compare new questions to Lesson 3 additions — assess whether the quality of questions is becoming more mechanistic and quantitative (a sign of deepening model sophistication). If groups only post repetitive questions, prompt: 'What does half-life tell us about WHERE on the N vs Z plot a nucleus will be found after many half-lives?' This connects back to the stability model built in Lessons 1 and 2 and reveals whether learners are integrating knowledge across lessons.
Model Building Phase 📹 **VIDEO:** [Worked example: Using oxidation numbers to identify oxidation and reduction] Source: Khan Academy (English - US curriculum)	Each group retrieves their cumulative nuclear stability model (the annotated N vs Z plot with decay arrows added in Lesson 2, and the detection-layer annotations added in Lesson 3). They now add a DECAY CURVE LAYER to their model. This consists of: (1) a small hand-drawn decay curve sketch pasted or drawn in the margin of the N vs Z	Distribute the cumulative models and say: 'Your model has been growing since Lesson 1. Today you are adding the TIME dimension — how fast does decay happen? Add a decay curve layer now. You have 6 minutes.' Circulate and prompt: 'Where on your N vs Z plot would a nucleus with a very SHORT half-life be found? Would it still be there after	Cumulative model building makes the progressive development of the explanatory framework visible and tangible. The peer-explanation protocol (90-second pitchblende explanation) checks whether learners can translate their model back to the anchoring phenomenon — the ultimate test of coherent understanding. Colour-coded half-life keys build intuition	Observe whether groups correctly orient their decay curve (exponential decrease, not linear); whether the half-life extraction dashed lines are placed accurately; and whether the annotation connecting uranium-238 to the pitchblende phenomenon is scientifically accurate (emphasising the immense half-life as the reason for

🔍 Search ARES for similar videos 📖 **HTML:** Nuclear Decay Processes (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	diagram, labelled with axes N (number of undecayed nuclei) and time, with the half-life marked by a dashed horizontal and vertical line; (2) an annotation arrow from uranium-238 on their N vs Z plot pointing to the decay curve, with the label 'half-life = 4.5 billion years — this is why the pitchblende rock still emits radiation today'; (3) a second annotation connecting Carbon-14 on the plot to the Olorgesailie dating application with the label 'half-life = 5730 years — used to date organic remains in Kenya'; (4) a colour-coded key distinguishing short half-life isotopes (red — dangerous, fast-decaying, used in medicine) from long half-life isotopes (blue — slow-decaying, geological timescale). Groups present their updated model to an adjacent group in a 90-second peer explanation. The receiving group provides one piece of feedback using the sentence frame: 'Your model explains ________ well, and one thing that could be clearer is ________.'	a million years?' If a group finishes early, extend: 'Add the Iodine-131 case to your model — where does it sit, what does its decay curve look like compared to uranium-238?' For the peer presentation, say: 'You have exactly 90 seconds. Your job is to use your model to EXPLAIN to the next group why the pitchblende rock from our opening phenomenon is still active today. Go.' After peer feedback, bring the class together and ask one group to share the most insightful piece of feedback they received. WAIT 8 seconds after asking this before accepting responses. Close by saying: 'Your model is now four lessons deep. In Lesson 5 we will add nuclear equations — the exact mathematical language for what happens inside the nucleus when it decays. Your model is almost complete.'	about the relationship between half-life, stability, and practical application.	persistence, not infinite nuclei). During peer presentations listen for explanations that incorporate probability language ('each nucleus has a chance of decaying') rather than deterministic language ('the nucleus decides to decay') — this distinction reflects genuine understanding of the statistical model built during the dice simulation.

D. TEACHER REFLECTION

After the lesson, consider: (1) Did the dice simulation make the concept of random-but-statistically-predictable decay genuinely clear, or did learners treat it as merely a game? Evidence: listen to the language in Model Building peer presentations — are learners using probabilistic vocabulary? (2) Were the two Kenya problems (Olorgesailie and Iodine-131) sufficiently grounded in context for learners to see their relevance, or did they feel like abstract substitutions? If the latter, consider opening Lesson 5 with a brief follow-up about what Olorgesailie reveals about early human occupation of the Rift Valley. (3) Were any groups unable to extract the half-life graphically — did they lack the graph-reading skill or the conceptual understanding? Plan a brief 5-minute remediation at the start of Lesson 5 using a pre-drawn decay curve if needed. (4) Did the DQB generate genuinely new questions, or were learners recycling earlier questions? New questions pointing toward nuclear equations or fission suggest readiness for Lesson 5. (5) Equity check: did all four roles in each group (roller, survivor-counter, decay-counter, recorder) rotate, ensuring all learners had hands-on access to the simulation rather than one dominant learner running the entire experiment?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	When we rolled 100 dice and removed every 6, the number of surviving dice decreased each round, producing a curved graph that flattened out over time rather than a straight line. Every group extracted a similar half-life of approximately 3 to 4 rolls despite different individual data patterns. We also observed that the number of dice removed per round decreased as the surviving population got smaller.
What did I learn?	The half-life of a radioactive isotope is the time for half the undecayed nuclei in a sample to decay. Decay follows an exponential pattern described by N = N0 x (one-half)^n. Different isotopes have vastly different half-lives: Carbon-14 has a half-life of 5730 years (used to date the Olorgesailie charcoal) and Iodine-131 has a half-life of 8 days (used in KNH thyroid treatment). Uranium-238 has a half-life of 4.5 billion years, which explains why the pitchblende rock in our phenomenon is still emitting radiation today.
How does this explain the phenomenon?	The pitchblende rock has been emitting radiation for billions of years without running out because uranium-238 has an enormously long half-life. Only a tiny fraction of its nuclei have decayed since Earth formed, leaving vast numbers of undecayed nuclei still present. The decay process is random at the level of individual nuclei but statistically predictable for large populations — exactly as our dice simulation showed. The rate of emission is not constant forever (it slowly decreases as nuclei are used up) but on any human timescale the rock appears to emit radiation indefinitely.

LESSON 5 (80 minutes): Nuclear Equations: Writing and Balancing Radioactive Decay	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to write and balance nuclear equations for alpha, beta, and gamma decay, and to connect each decay type to shifts toward the belt of stability on the N vs Z plot.
Knowledge	Learners know that: (1) nuclear equations must conserve nucleon number (mass number) and proton number (atomic number); (2) alpha decay reduces nucleon number by 4 and proton number by 2; (3) beta-minus decay increases proton number by 1 while nucleon number stays constant; (4) gamma emission changes neither nucleon number nor proton number; (5) each decay moves the daughter nucleus closer to the belt of stability on the N vs Z plot.
Skills	Learners are able to: (a) write balanced nuclear equations for α-decay (e.g., ${}^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He$), β-decay (e.g., ${}^{14}_{6}C \rightarrow {}^{14}_{7}N + {}^{0}_{-1}e$), and γ-emission; (b) verify conservation of nucleon number and proton number on both sides of a nuclear equation; (c) plot the daughter nucleus on the N vs Z graph and describe its movement relative to the belt of stability; (d) identify the decay type given an incomplete nuclear equation and determine the missing daughter or particle.
Attitudes	Learners appreciate that mathematical conservation laws govern nuclear processes; they show persistence in balancing complex equations and intellectual curiosity about how invisible nuclear-level changes produce measurable, real-world effects — including the exposure of photographic film by pitchblende ore in complete darkness.
Key Inquiry Question	What mathematical rules govern nuclear decay equations, and how does each type of decay shift a nucleus toward stability on the N vs Z plot?
Purpose in Storyline	In Lessons 1–4, learners established why some nuclei are unstable (neutron-to-proton ratio outside the belt of stability), identified the three types of radiation, explored detection methods, and examined penetration properties. Now in Lesson 5 — the fifth evidence-gathering lesson — learners move from qualitative description to quantitative representation: they write and balance the actual nuclear equations that describe what is happening inside the pitchblende ore, lesson by lesson tying the invisible mark on the photographic film to precise nuclear arithmetic. This lesson is the algebraic language layer of the anchoring phenomenon, enabling learners to predict exactly which daughter nucleus forms and to verify that each decay is indeed a step toward stability.
Safety Notes	No radioactive materials are used. All work is pencil-and-paper and simulation-based. Ensure any online simulation (e.g., PhET Nuclear Fission or equivalent offline ARES module) is pre-loaded on devices before the lesson. Remind learners that real uranium ore — like pitchblende found in some Kenyan geological surveys — must never be handled without specialist shielding; the equations they write today describe real processes that require real safety protocols.

B. LESSON OVERVIEW
Learners open by revisiting the anchoring phenomenon — the pitchblende ore exposing photographic film in darkness — and are challenged to move beyond naming the radiation types to writing the precise nuclear arithmetic that describes what is happening inside every uranium atom in that rock. Through a structured Predict–Observe–Explain cycle, learners first predict how particle ejection changes proton and nucleon numbers, then observe worked examples and an interactive simulation (or ARES offline equivalent) to gather evidence, then explain by writing and balancing their own nuclear equations for α-decay, β-decay, and γ-emission. A critical sensemaking moment connects each balanced equation back to the N vs Z stability plot from Lesson 1: learners physically plot the daughter nucleus and trace its movement toward the belt of stability, making the abstract equation visually meaningful. The lesson closes with Driving Question Board (DQB) updates and cumulative model revision, cementing nuclear equations as the quantitative core of the unit's explanatory model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Writing nuclear equations for alpha, beta, and gamma decay Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Conservation of Mass and Energy in Nuclear Reactions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a single prompt card showing the pitchblende phenomenon image (ore near photographic film in darkness) alongside a partially written nuclear equation: $^{238}_{92}U \rightarrow$??? + $^{4}_{2}He$. Working individually for 3 minutes in silence, they predict: (a) What numbers should replace the ??? for the nucleon number and proton number of the daughter nucleus? (b) Which element on the periodic table does that daughter nucleus belong to? (c) On a blank N vs Z grid (pre-printed on their worksheet), they mark where $^{238}_{92}U$ sits and draw an arrow showing where they think the daughter nucleus will land. Learners then share predictions with a shoulder partner for 2 minutes, noting agreements and disagreements in their science notebooks.	Distribute prompt cards and worksheets. SAY: 'Before I teach you anything about nuclear equations, I want you to use what you already know from our previous four lessons — isotopes, decay types, radiation properties — to make your best mathematical prediction. You have 3 minutes of silent thinking time. Go.' [Allow 3 full minutes of silence — do NOT circulate or answer questions during this window.] After 3 minutes: 'Now, turn to your shoulder partner. You have 2 minutes. Tell each other your predicted nucleon number and proton number for the daughter nucleus, and show each other your arrows on the N vs Z grid.' After 2 minutes, cold-call 3 pairs using the class register (rows 1, 3, and 5): 'Akinyi and Mwangi — what nucleon number did you predict for the daughter, and why?' [Wait 10 seconds.] 'Fatuma and Otieno — did you agree with them on the proton number?' [Wait 10 seconds.] 'Wanjiku and Kipchoge — where did your arrow point on the N vs Z grid?' [Wait 12 seconds.] Record all three predictions on the board under the heading 'OUR PREDICTIONS — check them later.' Do NOT confirm or correct at this stage.	Activation of Prior Knowledge + Anchored Prediction: By requiring learners to make a numerical prediction before instruction, the teacher activates their understanding of proton numbers, nucleon numbers, and the N vs Z plot (built in Lessons 1–4). The phenomenon image on the prompt card anchors the abstract arithmetic to the real rock. Disagreements surfaced during pair-share create productive cognitive conflict that motivates the Observe phase.	Observable indicator: During cold-calls, listen for whether learners apply conservation reasoning (e.g., 'nucleon number must go from 238 to 234 because the alpha particle takes 4') versus guessing randomly. Note on a clipboard which learners show conservation thinking and which do not — these learners will receive targeted scaffolding during the Explain phase. Worksheet arrows on the N vs Z grid reveal whether learners understand that alpha decay moves the nucleus down and to the left.
Observe Phase 📹 **VIDEO:** Writing nuclear equations for	Evidence Activity 1 — Conservation Law Demonstration (15 min): The teacher writes a fully worked α-decay equation on the board step by step, narrating each	Evidence Activity 1: Write $^{238}_{92}U \rightarrow$ $^{234}_{90}Th$ + $^{4}_{2}He$ on the board slowly, narrating: 'Watch the top numbers — nucleon numbers — on both sides. On the left: 238. On the	Worked Example → Guided Practice → Independent Simulation (I Do / We Do / You Do): This scaffolded sequence progressively transfers ownership	Observable indicators: (1) During trio work, check whether all three numbers (A, Z, and element symbol) are correctly filled in on all three cards — collect one card per

alpha, beta, and gamma decay Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Conservation of Mass and Energy in Nuclear Reactions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	move aloud. Learners copy into notebooks and annotate each number with its meaning (top number = nucleon/mass number A; bottom number = proton/atomic number Z). They check: does A balance? Does Z balance? They confirm using a periodic table strip (provided) that Z=90 is Thorium. Evidence Activity 2 — Guided Trio Practice (15 min): In groups of three, learners receive three equation cards: Card A (α-decay: $^{226}_{88}Ra \rightarrow$??? + $^{4}_{2}He$, Radium found in pitchblende ore alongside Uranium), Card B (β-minus decay: $^{14}_{6}C \rightarrow$??? + $^{0}_{-1}e$, Carbon-14 used in dating Kenyan archaeological sites at Fort Jesus, Mombasa), Card C (γ-emission: $^{60}_{27}Co \rightarrow {}^{60}_{27}Co^{*} + \gamma$, Cobalt-60 used in Kenya's cancer treatment machines at KNH Nairobi). Each group must: (i) fill in the missing daughter nucleus symbol, (ii) verify conservation on both sides, (iii) use a periodic table to name the daughter element. Evidence Activity 3 — ARES Simulation / Offline Interactive (10 min): Learners use the ARES offline simulation module (or a pre-downloaded PhET-equivalent) in which they input a parent nucleus and observe the daughter nucleus and emitted particle. They record three decay observations in a table: Parent → Daughter | ΔA | ΔZ | Decay type. This provides direct visual and interactive evidence of how numbers change.	right: 234 plus 4. What is 234 plus 4?' [Wait 8 seconds, then take choral response.] 'Exactly — 238. The top numbers balance. Now watch the bottom numbers — proton numbers. Left: 92. Right: 90 plus 2. Does that balance?' [Wait 8 seconds, choral response.] 'So the rule is: the SUM of nucleon numbers on the right must equal the nucleon number on the left. Same for proton numbers. This is called conservation of nucleon number and conservation of proton number. Write that rule in a box in your notebook.' Circulate during trio practice — spend no more than 90 seconds per group. Use prompting questions: 'What does the top number on the alpha particle tell you?' / 'What Z number does your answer give you — look it up on the periodic table strip.' For Card C (gamma): SAY to the class: 'Gamma emission is the special case — check your numbers. What changes?' [Wait 10 seconds.] Cold-call one group: 'Chebet's group — what did you find changes in gamma emission?' [Wait 10 seconds.] During the simulation phase, circulate and ask: 'What happens to the neutron number — N — when alpha decay occurs? Is the daughter above or below the parent on the N vs Z grid?'	of nuclear equation balancing from teacher to learner. The three equation cards use Kenya-relevant contexts (pitchblende ore, Fort Jesus carbon dating, KNH cancer treatment) so each example is simultaneously a piece of evidence about the anchoring phenomenon and real-world nuclear science. The simulation provides dynamic, visual evidence that numbers change predictably — turning an abstract rule into an observable pattern.	group at the end as an exit check. (2) During simulation, learners' recorded tables should show consistent ΔA = −4, ΔZ = −2 for α-decay and ΔA = 0, ΔZ = +1 for β-decay. Spot-check three tables mid-simulation. (3) Listen for the response to 'What changes in gamma emission?' — the correct answer (nothing changes in A or Z) reveals conceptual understanding beyond rote rule application.
Explain Phase 📹 **VIDEO:** Writing nuclear equations for alpha, beta, and gamma decay	Learners return individually to the N vs Z stability plot printed on their worksheet. They have already plotted $^{238}_{92}U$ (from the phenomenon) in Lesson 1. Now they: (1) Plot the daughter nucleus	SAY: 'You have now gathered enough evidence to explain the phenomenon at the nuclear arithmetic level. Pull out the N vs Z grid on your worksheet. You have 12 minutes to complete steps 1	Claim–Evidence–Reasoning (CER) Writing + N vs Z Visual Mapping: Learners construct a written CER explanation (Claim: the ore exposes film because uranium nuclei decay; Evidence:	Observable indicators: (1) N vs Z arrows must point in the correct direction — α-decay arrow goes down-left (ΔA=−4, ΔZ=−2); β-decay arrow goes right (ΔZ=+1, ΔA=0). Incorrect arrow direction

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Conservation of Mass and Energy in Nuclear Reactions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	$^{234}_{90}Th$ after α-decay and draw an arrow from parent to daughter, labelling it 'α-decay.' (2) Assume $^{234}_{90}Th$ then undergoes β-minus decay — they write the balanced equation in their notebook ($^{234}_{90}Th \rightarrow {}^{234}_{91}Pa + {}^{0}_{-1}e$), plot $^{234}_{91}Pa$, and draw a second arrow labelled 'β-decay.' (3) They annotate each arrow: 'Moves toward the belt of stability because...' completing the sentence in their own words. (4) They answer a connecting question in writing: 'Using your nuclear equations, explain in 3–4 sentences exactly what is happening inside the pitchblende ore that caused the photographic film to be exposed in complete darkness.' Learners share their written explanations with a new partner (not their trio partner) and give each other one piece of feedback using the sentence stem: 'Your explanation would be stronger if you also mentioned...'	through 4 individually — this is your sensemaking time, work in silence.' [Set a visible 12-minute timer.] Circulate purposefully — target learners flagged during Phase 1 as lacking conservation reasoning. Use the prompting question: 'Your arrow for alpha decay — which direction does it go on the grid? Is the daughter nucleus closer to or farther from the belt of stability?' After 12 minutes: 'Find a partner who is NOT from your trio group. Read each other's written explanation for the phenomenon. Give one piece of feedback using the sentence stem on the board: Your explanation would be stronger if you also mentioned...' [Allow 5 minutes for peer feedback.] Cold-call 2 learners to read their revised explanations aloud: 'Njoroge — read us your explanation of what is happening inside the pitchblende ore.' [Wait 10 seconds.] 'Achieng — what did your partner suggest you add, and do you agree?' [Wait 12 seconds.] Affirm correct use of: nucleon number, proton number, conservation, daughter nucleus, belt of stability. Correct misconceptions immediately: if a learner says 'the atom loses energy' redirect: 'Be more precise — which specific particles or energy does the nucleus eject, and what are the exact numbers?'	balanced nuclear equations showing specific particle ejection; Reasoning: conservation laws confirm the identity of emitted particles and daughter nuclei). Simultaneously, plotting decay arrows on the N vs Z grid makes the abstract equation geometrically meaningful — learners see stability-seeking as a physical journey on the graph, not just an algebraic exercise. Peer feedback builds scientific communication skills.	reveals a conceptual gap. (2) Written phenomenon explanations must mention: unstable nucleus, specific emitted particle (alpha or beta), conservation of nucleon and proton numbers, daughter nucleus moving toward belt of stability. Collect written explanations as an embedded assessment artifact. (3) Peer feedback quality — listen for whether learners are offering substantive scientific feedback vs generic comments.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Writing nuclear equations for alpha, beta, and gamma decay	Each learner receives two sticky notes. On Note 1 (green): they write one thing their nuclear equations now allow them to answer from the Driving Question Board — 'Why do some atoms in Kenyan rocks spontaneously release invisible energy?' — using specific language from today's	Distribute sticky notes. SAY: 'You have 4 minutes to write your two notes. Green note: what can you now answer about the driving question, using today's specific vocabulary? Orange note: what new question has nuclear equations raised for you?' [Set 4-minute timer. Circulate silently.]	Driving Question Board Curation: The DQB functions as a public, cumulative record of the class's growing explanatory model. Green notes make learners articulate what the lesson has resolved; orange notes maintain productive intellectual tension by surfacing what remains unexplained. The	Observable indicators: (1) Green notes should reference specific nuclear vocabulary from today — any note that only says 'radiation' without mentioning conservation, nucleon number, or daughter nucleus indicates surface-level understanding. (2) Orange notes reveal anticipatory thinking —

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Conservation of Mass and Energy in Nuclear Reactions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	lesson (nucleon number, proton number, conservation, daughter nucleus). On Note 2 (orange): they write one new question that today's lesson has raised for them — for example, about what happens after many successive decays, or how fast these decays occur. Learners post both notes on the class DQB. The teacher facilitates a 5-minute whole-class gallery walk where learners read three peers' green notes and give a silent thumbs-up to one they find most complete and precise. The teacher then selects 2 green notes and 2 orange notes to read aloud, highlighting the language of conservation and stability.	After posting: 'Stand up. Do a silent gallery walk — read at least three green notes from people who are NOT in your trio group. Give a silent thumbs-up to the one you think is most scientifically complete. You have 3 minutes.' [After gallery walk:] Read two green notes aloud — choose one strong example and one partial example. For the partial example: 'This note says the uranium atom releases energy. That is true — but using today's lesson, how could we make this more precise?' [Wait 10 seconds, cold-call: 'Kamau — how would you improve this note?'] [Wait 10 seconds.] Then read two orange notes: 'These are excellent new questions — notice that Lesson 6 will begin to answer this one about rate of decay. Keep these questions in mind as we go forward.' Update the DQB column labelled 'Lesson 5 — Nuclear Equations' with selected notes.	gallery walk creates a low-stakes peer-assessment environment where learners calibrate their own understanding against their classmates'. Teacher selection of notes for public reading models the scientific norm of precise, evidence-based language.	questions about decay rate, decay chains, or energy released suggest readiness for Lessons 6 (half-life) and 7 (applications/fission/fusion). (3) The thumbs-up distribution reveals class consensus on what a strong explanation looks like — if thumbs cluster on vague notes, a brief whole-class norm-setting moment is needed.
Model Building Phase 📹 **VIDEO:** Writing nuclear equations for alpha, beta, and gamma decay Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Conservation of Mass and Energy in Nuclear Reactions (Flexbook) Source: CK-12	Learners open the cumulative 'Pitchblende Phenomenon Model' they have been building since Lesson 1 (a hand-drawn annotated diagram in their science notebook). Today they add a new layer: a 'Nuclear Equation Panel' adjacent to their diagram of the uranium nucleus. In this panel they: (1) Write the full balanced equation for $^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He$, annotating each symbol (A, Z, element symbol, particle name). (2) Draw a small arrow from their uranium nucleus diagram pointing to the alpha particle and the thorium daughter, labelling each with its A and Z values. (3) Write a one-sentence model statement: 'When a $^{238}_{92}U$ nucleus in	SAY: 'Open your Pitchblende Phenomenon Model from Lesson 1. Today you are adding the nuclear equation layer — the precise mathematical description of what happens at the moment a uranium nucleus in that rock releases invisible radiation. You have 8 minutes. Add your Nuclear Equation Panel, your annotated arrow, and your model statement. If you finish early, add the next decay step.' [Set 8-minute timer. Circulate — specifically check that A and Z annotations are correct and that the model statement connects explicitly to both the equation AND the N vs Z plot.] With 2 minutes remaining: 'Add your model limitation note — what	Cumulative Model Revision (Progressive Refinement): The Pitchblende Phenomenon Model grows lesson by lesson, making visible the accumulation of explanatory depth. Adding a 'Nuclear Equation Panel' to an existing diagram — rather than starting fresh — reinforces that science builds incrementally. The 'Model Limitation' note is a deliberate metacognitive move: it teaches learners that a good scientific model acknowledges its own boundaries, and it creates anticipatory readiness for Lesson 6. The optional decay chain extension challenges fast finishers while foreshadowing the uranium decay series concept.	Observable indicators: (1) The Nuclear Equation Panel must show a correctly balanced equation with A and Z values annotated — missing or incorrect annotations are a formative flag. (2) The model statement must contain: the parent nucleus (with A and Z), the emitted particle, the daughter nucleus (with A and Z), and a reference to the belt of stability. Statements missing the belt of stability reference show that learners have not yet connected the equation to the stability framework. (3) The model limitation note reveals metacognitive awareness — a note that says 'this only shows one step' is more sophisticated than a

🔍 Search ARES for similar readings	pitchblende ore ejects an alpha particle, it becomes $^{234}_{90}Th$ — a daughter nucleus with nucleon number 234 and proton number 90 — moving closer to the belt of stability on the N vs Z plot.' (4) Add a 'Model Limitation' note: 'This model shows one decay step. The actual decay chain of uranium involves many successive decays — our model will grow in Lessons 6 and 7.' Learners who finish early add the β-decay step ($^{234}_{90}Th \rightarrow {}^{234}_{91}Pa$) to their model, beginning to sketch a decay chain.	does this model NOT yet explain?' [After time:] Cold-call one learner to share their model statement: 'Wambui — read us your one-sentence model statement.' [Wait 10 seconds.] Affirm the connection to the phenomenon explicitly: 'Notice what Wambui has done — she has used a balanced nuclear equation to explain exactly what produces the invisible radiation that exposed the photographic film. That is the power of nuclear equations: they turn an invisible mystery into precise, verifiable mathematics.' Close by previewing Lesson 6: 'Your model now explains WHAT is emitted and HOW the nucleus changes. In Lesson 6, we will ask: HOW FAST does this happen? That is the concept of half-life — and it will help us answer how long Kenyan uranium ore will remain radioactive.'		blank or 'I don't know.' Photograph two or three model pages with a tablet to share as examples in the next lesson's opening.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. CONSERVATION FLUENCY: During Phase 2 trio practice, which groups consistently applied conservation of nucleon number and proton number correctly from the first attempt, and which groups required prompting? What scaffolding (e.g., a 'check-sum box' template) would help struggling learners in a re-teach moment?

2. N vs Z ARROWS: In Phase 3, did learners draw their decay arrows in the correct direction on the N vs Z grid? Specifically — did any learners draw the α-decay arrow going UP-right instead of DOWN-left? This is a persistent misconception (confusing 'gaining mass' with 'gaining stability'). Plan a targeted 5-minute mini-lesson if more than one-third of the class made this error.

3. PHENOMENON CONNECTION: When cold-called in Phases 3 and 5, did learners spontaneously reference the pitchblende ore and the photographic film in their explanations, or did they explain decay in abstract terms only? If the phenomenon connection was weak, build in a deliberate 'bring it back to the rock' prompt at the start of every future lesson phase.

4. KENYA CONTEXT ENGAGEMENT: Did the three equation cards (pitchblende/Radium, Fort Jesus/Carbon-14, KNH Cobalt-60) generate genuine curiosity or feel forced? Note which context sparked the most questions from learners — use that context to open Lesson 6 (half-life).

5. DQB QUALITY: Review the green and orange sticky notes posted on the DQB. Do the green notes use precise nuclear vocabulary (nucleon number, proton number, daughter nucleus, belt of stability)? If the majority of green notes are vague ('uranium releases radiation'), plan a vocabulary reinforcement activity at the start of Lesson 6. Do the orange notes anticipate half-life or decay chains? If so, use these learner-generated questions as the opening hook for Lesson 6.

6. MODEL BUILDING: Were learners' 'Nuclear Equation Panels' annotated with A and Z values, or did learners copy the equation without understanding what each symbol means? If annotation was weak, introduce a 'symbol decoding' warm-up routine for Lessons 6 and 7.

7. PACING: This lesson is ambitious for 80 minutes. Note which phase overran. If Phase 2 (Observe) consistently takes longer than 40 minutes, consider splitting Evidence Activity 3 (simulation) into an independent homework task using the ARES offline module, and use the recovered time to deepen Phase 3 (Explain) peer feedback.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed: (1) a fully worked α-decay equation (${}^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He$) demonstrated step-by-step on the board, showing conservation of nucleon number and proton number; (2) three guided equation cards covering α-decay (Radium in pitchblende), β-decay (Carbon-14 dating at Fort Jesus, Mombasa), and γ-emission (Cobalt-60 at KNH Nairobi), each requiring the daughter nucleus to be identified using a periodic table; (3) an interactive simulation (ARES offline module) in which parent nuclei were input and daughter nuclei and emitted particles appeared, with ΔA and ΔZ values recorded in a table.
What did I learn?	Learners learned that: (1) nuclear equations must conserve nucleon number (A) and proton number (Z) on both sides — these are the fundamental rules governing all radioactive decay; (2) α-decay always reduces A by 4 and Z by 2, producing a daughter nucleus two elements lower on the periodic table; (3) β-minus decay keeps A constant while increasing Z by 1, producing a daughter nucleus one element higher; (4) γ-emission changes neither A nor Z — it releases energy only; (5) each decay can be plotted as a directional arrow on the N vs Z graph, and every arrow points toward the belt of stability, confirming that radioactive decay is the nucleus's way of achieving a more stable configuration.
How does this explain the phenomenon?	Learners explained: Using balanced nuclear equations and the N vs Z stability plot, learners explained that the pitchblende ore exposes photographic film in complete darkness because ${}^{238}_{92}U$ nuclei — sitting outside the belt of stability — spontaneously eject alpha particles (${}^{4}_{2}He$), transforming into ${}^{234}_{90}Th$ daughter nuclei that sit closer to the belt of stability. The emitted alpha particles (and subsequent beta particles and gamma rays from further decays in the chain) carry enough energy to penetrate the dark container and interact with the photographic film's silver halide crystals, leaving a visible exposure. Conservation of nucleon number and proton number governs every step, making the invisible process mathematically predictable and verifiable.

LESSON 6 (80 minutes): Nuclear Fission and Fusion: Unlocking the Energy Inside the Rock	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate how nuclear fission and fusion release enormous amounts of energy, connect this to the energy source inside the anchoring phenomenon rock, and construct evidence-based arguments about Kenya's nuclear energy future.
Knowledge	Learners will be able to: (a) describe nuclear fission as the splitting of a heavy nucleus (e.g., uranium-235) into smaller nuclei, releasing energy and neutrons; (b) describe nuclear fusion as the joining of light nuclei (e.g., hydrogen isotopes) to form a heavier nucleus, releasing even greater energy; (c) explain mass defect and binding energy conceptually — that the 'missing' mass is converted to energy via $E = mc^2$; (d) distinguish between a chain reaction in a nuclear reactor and in a nuclear weapon; (e) outline the role of Kenya Nuclear Electricity Board (KNEB) in Kenya's energy planning.
Skills	Learners will be able to: (a) use the PhET Nuclear Fission simulation to gather evidence about chain reactions and energy release; (b) construct and defend an evidence-based argument about whether Kenya should pursue nuclear power; (c) write and balance a simple nuclear fission equation; (d) compare fission and fusion using a structured comparison table; (e) update the class Driving Question Board with new evidence.
Attitudes	Learners will: (a) appreciate that the enormous energy locked in atomic nuclei — first hinted at by the glowing pitchblende rock — has the potential to transform Kenya's energy landscape; (b) demonstrate responsible thinking by weighing benefits and risks of nuclear energy for Kenya; (c) show intellectual honesty by acknowledging uncertainty and complexity in real-world energy decisions.
Key Inquiry Question	How does splitting or joining atomic nuclei release such enormous energy — and could that same energy that makes our pitchblende rock 'dangerous' also power Kenyan homes and hospitals?
Purpose in Storyline	In Lessons 1–5, learners established that the pitchblende rock's uranium nuclei are unstable, that they decay by emitting α, β, and γ radiation, that decay follows a predictable half-life, and that radiation can be detected and shielded. But one question has lingered on the DQB: 'Where does all that energy ACTUALLY come from?' Lesson 6 answers this directly — the energy comes from the conversion of a tiny amount of nuclear mass into energy ($E = mc^2$) during nuclear rearrangement. Learners then leap forward: if natural radioactive decay releases energy slowly, what if we deliberately split (fission) or join (fusion) nuclei to release that energy all at once? This lesson positions nuclear energy as the technological harnessing of the same phenomenon they have been studying, and forces them to grapple with Kenya's real decision about nuclear power (KNEB's planned 1,000 MW reactor).
Safety Notes	No radioactive materials are used in this lesson. The PhET simulation is digital and entirely safe. When watching video content, ensure volume levels are appropriate. During the debate activity, remind learners that this is an academic argument — personal attacks are not permitted. Teacher should clarify that nuclear reactors and nuclear weapons operate on different principles and that Kenya's nuclear programme is civilian and regulated.

B. LESSON OVERVIEW

Lesson 6 of 7 in the radioactivity evidence-gathering sequence. Learners open by revisiting the anchoring phenomenon — the pitchblende rock has been releasing energy for billions of years without any external input. Today they finally answer WHERE that energy comes from. Using the PhET Nuclear Fission simulation, a curated YouTube explainer (IAEA or Kurzgesagt-style), and structured discussion, learners investigate how uranium-235 undergoes fission (splitting), how hydrogen nuclei undergo fusion (joining), and how mass defect explains the energy released in both cases. They then apply this to Kenya's energy context: KNEB's plans for a 1,000 MW nuclear power plant, comparing nuclear energy to Kenya's existing geothermal (Olkaria), hydro (Turkwel), and solar sources. The lesson culminates in a structured

academic controversy debate — 'Should Kenya build a nuclear power plant?' — in which learners construct evidence-based arguments, and the DQB is updated with new understanding. A cumulative nuclear model is revised to show fission and fusion pathways alongside the natural decay pathway studied in earlier lessons.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Bacteria Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Conservation of Mass and Energy in Nuclear Reactions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a projected image of two scenarios side by side: (A) the pitchblende rock sitting on a table slowly fogging a photographic film (the anchoring phenomenon), and (B) a nuclear power plant cooling tower in the background with lights blazing across a city at night. The teacher poses the question: 'Both of these involve uranium. In Scenario A, the rock quietly releases a little energy over billions of years. In Scenario B, a power plant produces enough electricity to light up Nairobi. What must be happening DIFFERENTLY in the power plant?' Learners individually write a 3-sentence prediction in their science notebooks, then share with a shoulder partner. Selected pairs share with the class. The teacher records key prediction phrases on the board without evaluating them — they will be revisited at the end of the lesson.	1. Project the split image and say: 'Take 60 seconds — look carefully at both images. What is the same? What must be different? Write three sentences predicting what is happening differently in the nuclear power plant compared to our pitchblende.' [Allow 60 seconds silent writing.] 2. Say: 'Now turn to your shoulder partner and share your prediction. You each have 45 seconds.' [Allow 90 seconds pair talk.] 3. Cold-call 4 pairs (front-left, back-right, front-right, back-left): 'Pair at the front left — what was your prediction?' After each response, say ONLY: 'Interesting — let me write that up.' Do NOT correct or affirm yet. 4. Record 4–6 student prediction phrases verbatim on a 'Prediction Wall' section of the board. Circle any that mention 'splitting atoms,' 'chain reaction,' 'more uranium,' or 'controlled explosion.' 5. Say: 'Hold onto these predictions. By the end of today, we will come back and see who was closest — and more importantly, WHY.'	Predict-then-Revisit: By making explicit predictions before instruction, learners activate prior knowledge (decay, instability, energy) and create cognitive dissonance — a gap between what they think they know and what they will discover. This motivates engagement with the simulation and video evidence.	Observable indicator: All learners produce a written 3-sentence prediction within 60 seconds. Teacher scans notebooks during pair-share for evidence of prior knowledge activation — look for: (a) references to 'splitting' or 'joining' atoms (strong prior knowledge), (b) references to 'more atoms/more uranium' (common misconception to address), (c) blank or off-topic responses (flag these learners for targeted questioning during Phase 2). Teacher notes the 2–3 most productive misconceptions on a sticky note to address explicitly in Phase 3.
Observe Phase 📹 **VIDEO:** Bacteria Source: Khan Academy (English - US curriculum)	The Observe phase has two sequential evidence-gathering activities: ACTIVITY A — PhET Nuclear Fission Simulation (25 minutes): In	ACTIVITY A SETUP (3 minutes before simulation): 1. Say: 'Before you open the simulation, I want you to remember our rock. Inside that pitchblende, uranium nuclei are	Structured Inquiry with Simulation + Two Facts / One Question: The PhET simulation allows learners to manipulate variables (add neutrons, add control rods) in a safe, visual environment —	Observable indicators: (a) Every learner's notebook contains 4 sketches from the simulation — teacher spot-checks 6 notebooks during circulation, flagging any that are missing the chain reaction

🔍 Search ARES for similar videos 📖 **HTML:** Conservation of Mass and Energy in Nuclear Reactions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	pairs sharing one device (or groups of 3 if devices are limited), learners open the PhET 'Nuclear Fission' simulation (pre-loaded offline on ARES tablets or accessed via https://phet.colorado.edu). They work through a structured investigation guide with four steps: Step 1 — Fire a single neutron at a U-235 nucleus. Observe what happens. Sketch the before-and-after in their notebooks and record: 'What products were released?' Step 2 — Enable 'Chain Reaction' mode. Fire one neutron and observe. Record: 'How many fissions happened from 1 neutron? How much energy was shown?' Sketch the branching chain. Step 3 — Add control rods. Observe how the chain reaction changes. Record: 'What do control rods do? Why is this important in a power plant?' Step 4 — Switch to the 'Nuclear Reactor' tab. Observe how the sustained chain reaction heats water to produce steam and turn a turbine. Sketch the energy transformation chain: nuclear energy → thermal energy → kinetic energy → electrical energy. ACTIVITY B — Video Explainer (10 minutes): Learners watch a 6-minute curated video segment (e.g., IAEA 'How Does a Nuclear Power Plant Work?' or a teacher-prepared Kenyan-context explainer) focusing on: (i) the difference between fission and fusion, (ii) what mass defect means conceptually ($E = mc^2$), and (iii) one slide about KNEB and Kenya's nuclear ambitions.	decaying slowly and naturally. Today we ask: what if we made a uranium nucleus split DELIBERATELY and RAPIDLY? Open PhET Nuclear Fission now.' 2. Distribute the printed (or digital) 4-step investigation guide. Say: 'You have 22 minutes for the simulation. Work in your pairs. BOTH partners must sketch every diagram — not just one person. I will be checking notebooks.' 3. CIRCULATE during simulation: After 5 minutes, pause the class for 30 seconds: 'Hands up if you have completed Step 1.' [Check.] 'What did you observe coming OUT of the nucleus after the split?' Cold-call 3 groups. Expected answers: 2–3 neutrons, energy, two smaller nuclei (fission fragments). Say: 'Those neutrons are KEY — hold that thought for Step 2.' 4. After Step 2, pause again: 'Who can describe what happened in the chain reaction in ONE sentence?' Wait 10 seconds. Cold-call 2 groups. Affirm: 'Yes — one neutron triggered a cascade. That cascade is called a CHAIN REACTION. Write that term in your notebooks now.' 5. After Step 3, ask: 'Why would a power plant NEED control rods? Discuss with your partner for 30 seconds.' [30 seconds.] Cold-call 2 pairs. ACTIVITY B SETUP: 6. Say: 'Now we zoom out. The simulation showed fission — splitting. But there is another nuclear reaction that releases even MORE energy. Watch the video. Your job: write down exactly	making the invisible nuclear world observable. The structured 4-step guide prevents aimless clicking and ensures every learner gathers the same core evidence. The 'Two Facts and One Question' protocol during the video activates metacognition — learners must distinguish what they now know from what they still wonder, directly feeding the DQB in Phase 4.	sketch. (b) During the two mid-simulation cold-call pauses, teacher listens for correct use of 'fission fragments,' 'neutrons,' and 'chain reaction' — if fewer than 60% of cold-called groups use correct terms, teacher pauses the class for a 2-minute whole-class clarification before proceeding to Step 3. (c) Video exit slip: 'Two Facts and One Question' — teacher scans slips as they are written; a correct 'fact' must reference either fission, fusion, mass defect, or KNEB. Any slip with two incorrect facts triggers a 1-on-1 clarifying question to that learner at the start of Phase 3.

	Learners complete a 'Two Facts and One Question' exit slip during the video.	TWO facts you learned and ONE question you still have.' [Play video.] 7. After video: 'Turn to your partner — share your one question.' [45 seconds.] Collect 5–6 questions on the board. Star any that mention fusion, mass defect, or safety — these will be addressed in Phase 3.		
Explain Phase 📹 **VIDEO:** Bacteria Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Conservation of Mass and Energy in Nuclear Reactions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The Explain phase has three parts: PART A — Fission vs. Fusion Comparison Table (10 minutes): Learners individually complete a structured comparison table in their notebooks with columns: 'Fission' vs. 'Fusion,' and rows: 'What nuclei are involved?', 'What happens?', 'Energy released (relative)?', 'Where does this happen naturally?', 'Human application,' 'Main challenge/risk.' They use evidence from the simulation, the video, and the teacher's explanation of mass defect. PART B — Mass Defect and E = mc² (Conceptual, 8 minutes): The teacher uses a 'Ugali analogy' to explain mass defect conceptually: 'Imagine you have 1 kg of ugali flour and 0.5 kg of water. You combine them and cook. The ugali you get out weighs slightly LESS than the combined ingredients — where did that mass go? It was released as heat and steam. In nuclear reactions, the products weigh SLIGHTLY less than the reactants — and that tiny lost mass becomes an ENORMOUS amount of energy, because energy = mass × the speed of light SQUARED. The speed of light is	PART A: 1. Say: 'Using everything from the simulation and the video, complete this comparison table individually — 8 minutes, silent work.' [Circulate, observe, do NOT answer content questions yet — redirect with: 'Check your simulation sketches and your Two Facts slip.'] 2. After 8 minutes: 'Check your table against your partner's. Where do you disagree? Put a question mark in those cells.' [2 minutes pair check.] 3. Cold-call one pair per row of the table (6 pairs total) to share their answer. Correct and annotate on the class projected table. PART B: 4. Say: 'Now I am going to explain WHERE the energy actually comes from — and I want you to connect this back to our rock.' Deliver the ugali analogy (scripted above). Pause after 'colossal energy' and ask: 'So when our pitchblende rock emits an alpha particle — is the alpha particle + the new nucleus heavier or lighter than the original uranium nucleus?' [WAIT 12 seconds.] Cold-call 3 learners. Expected answer: lighter. Say: 'Exactly. The missing mass	Structured Academic Controversy (SAC) + Ugali Analogy: The SAC framework requires learners to argue a position they may not personally hold, which forces deep engagement with evidence on both sides — this is a hallmark of scientific thinking. The ugali analogy uses a familiar Kenyan cooking context to make the abstract concept of mass defect tangible and culturally grounded. The combination ensures that conceptual understanding (mass defect) is immediately applied in a real-world decision-making context (Kenya's energy future), closing the loop between nuclear physics and human relevance.	Observable indicators: (a) Comparison table — teacher collects 6 random notebooks at the end of Part A and checks that the 'Energy released' row correctly identifies fusion as releasing more energy than fission, and that 'human application' rows are correctly populated. Any learner with more than 3 empty or incorrect cells is flagged for Lesson 7 support. (b) Mass defect understanding — after the ugali analogy, the cold-call response to 'heavier or lighter?' is used as a class-level check. If more than 3 of the 3 cold-called learners answer incorrectly, the teacher re-teaches using the board before moving to the debate. (c) Debate evidence use — teacher listens specifically for whether each group cites at least one piece of numerical evidence from the cards. Groups that argue purely from opinion (no data) are asked at the synthesis stage: 'What evidence from the cards would support that claim?' This is recorded as a participation/reasoning indicator.

	300,000,000 metres per second — squared. Even a tiny mass becomes colossal energy.' Learners annotate this analogy in their notebooks and write $E = mc^2$ with a verbal definition. PART C — Kenya Nuclear Debate: Structured Academic Controversy (15 minutes): The class is divided into four groups of roughly equal size. Groups 1 & 2 are assigned PRO-nuclear (Kenya SHOULD build a nuclear power plant). Groups 3 & 4 are assigned ANTI-nuclear (Kenya SHOULD NOT build a nuclear power plant yet). Each group has 5 minutes to build their argument using an evidence card pack (teacher-prepared cards with real data: Kenya's current energy mix, KNEB timeline, cost comparisons, Chernobyl/Fukushima summaries, Kenya's geothermal success at Olkaria, uranium deposits in Kwale County). Groups then present their strongest argument in 90 seconds each. After all four presentations, the teacher facilitates a 3-minute 'synthesis' — learners who were PRO must now find ONE valid point from the ANTI side, and vice versa. The teacher closes by saying: 'A scientist's job is not to win the argument — it is to follow the evidence. Hold your position lightly.'	became the kinetic energy of the alpha particle and the gamma ray. THAT is where the energy in our rock comes from. Write that sentence now.' PART C — DEBATE SETUP: 5. Assign groups, distribute evidence card packs. Say: 'You have 5 minutes to build your argument. Use the evidence cards — not just your opinions. Your argument must include at least ONE piece of numerical evidence.' 6. During group prep, circulate and whisper to struggling groups: 'What does the cost-per-megawatt card tell you? How does that support your side?' 7. Run presentations: strictly time each group at 90 seconds. After all four: 'Now — PRO groups, I want you to tell me the STRONGEST point the anti-nuclear groups made. ANTI groups, tell me the strongest pro-nuclear point.' [WAIT 10 seconds after posing question.] Cold-call 2 groups from each side. 8. Close with: 'Scientists and policymakers in Kenya — at KNEB, at the Energy Ministry, at universities in Nairobi — are having exactly this debate right now. Your job today was to practise thinking like them: using evidence, acknowledging trade-offs, and being honest about what you do not yet know.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Bacteria] Source: Khan Academy (English	Learners return to the class Driving Question Board, which has been accumulating sticky notes and string connections since Lesson 1. The DQB displays the central driving question: 'Why do some atoms in Kenyan rocks	1. Direct learners' attention to the DQB. Say: 'Look at our board. Since Lesson 1, we have been building up a picture of what is happening inside this rock. Today we answered a question that has been sitting on our board since	Driving Question Board with Colour-Coded Sticky Notes and Visible Connections: The two-colour protocol (green = new understanding, yellow = new question) separates knowledge consolidation from inquiry	Observable indicators: (a) Every learner produces two sticky notes — any learner who has not written both is approached by the teacher with the prompt: 'Tell me one thing — just one — that surprised you today.' This oral response is

- US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Conservation of Mass and Energy in Nuclear Reactions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	spontaneously release invisible energy — and how can we harness or protect ourselves from that energy?' Today's DQB update focuses specifically on the word 'HARNESS' in the driving question — for the first time in the sequence, learners have direct evidence for how the energy in rocks like pitchblende can be deliberately harnessed at scale. Each learner writes TWO sticky notes: one GREEN (something they now understand that they did not understand at the start of today) and one YELLOW (a question that today's lesson has opened up or made more complex). Learners physically place their notes on the DQB. The teacher reads 5–6 notes aloud and uses string/arrows to connect new understanding to earlier concepts on the board (e.g., connecting today's 'chain reaction' note to Lesson 3's 'alpha decay releases energy' note, and to Lesson 5's 'half-life of U-235 is 700 million years' note).	Lesson 1 — [point to the question]: WHERE DOES THE ENERGY COME FROM? What is the answer now?' [WAIT 10 seconds.] Cold-call 3 learners. Expect: mass defect / $E = mc^2$ / tiny mass converted to huge energy. 2. Say: 'Now — write your two sticky notes. GREEN: one thing you understand now. YELLOW: one new question today's lesson opened up. You have 3 minutes.' [Allow 3 minutes silent writing.] 3. As learners place notes, teacher reads 5–6 aloud: 'Listen to what your classmates are thinking...' For each green note, teacher asks: 'How does this connect to what we learned in an earlier lesson?' For each yellow note, teacher says: 'Hold that question — it may be answered in Lesson 7, or it may be a question that scientists are STILL working on.' 4. Use string or a marker line to visibly connect: 'nuclear fission releases energy' → 'mass defect' → 'alpha/beta decay also converts mass to energy (Lesson 2)' → 'half-life of U-235 means fuel lasts a long time (Lesson 5).' Say: 'Look at this web. This is how science works — each piece of evidence connects to others. Our rock is not just a curiosity anymore. It is a window into one of the most powerful forces in nature.'	extension. Making the connections visible with string/lines externalises the learner's growing conceptual web — a key sensemaking move that shows learners that understanding is cumulative and connected, not a list of isolated facts. Publicly reading notes aloud validates individual thinking and creates a shared class understanding.	accepted in place of a written note and noted by the teacher. (b) The quality of green notes is assessed by the teacher in real time: a high-quality green note makes a specific connection to prior learning (e.g., 'I now understand that the energy in alpha particles — from Lesson 2 — comes from mass defect'). Low-quality notes (e.g., 'I learned about nuclear fission') are flagged and the teacher asks a follow-up probe question to that learner. (c) Yellow questions are scanned for persistence of the common misconception that 'nuclear reactors can explode like bombs' — if 3 or more yellow notes raise this, the teacher addresses it explicitly as a whole-class clarification before closing Phase 4.
Model Building Phase 📹 **VIDEO:** Bacteria Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners open their cumulative nuclear model — a diagram they have been building and revising since Lesson 1, which currently shows: a uranium nucleus (Lesson 1), the belt of stability and neutron-proton ratio (Lesson 1–2), alpha, beta, and gamma decay pathways with equations (Lessons 2–3),	1. Say: 'Take out your cumulative nuclear model. Pick up your red pen — today's additions go in red, so we can SEE what today's lesson added.' [Allow learners to locate their models — 1 minute.] 2. Project a simplified class version of the model on the board (the teacher's own version). Say: 'I	Cumulative Model Revision with Colour-Coded Layering: By using a different colour for each lesson's additions, learners can literally see the growth of their understanding over the 7-lesson sequence. The model is not a finished product but a living scientific model — exactly as scientists use models. The	Observable indicators: (a) Model completeness — teacher circulates during model building and stamps or ticks any model that correctly contains all four required additions (fission pathway, fusion pathway, $E = mc^2$ annotation, applications branch). Any model missing more than one component

📖 **HTML:** [Conservation of Mass and Energy in Nuclear Reactions (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	radiation detection symbols (Lesson 4), and a decay curve showing half-life (Lesson 5). Today, learners add a NEW BRANCH to their model specifically for 'deliberate/engineered nuclear reactions.' Using a coloured pen (a different colour from previous lessons — e.g., red for engineered reactions vs. blue for natural decay), they add: (a) a FISSION PATHWAY — a U-235 nucleus struck by a neutron, splitting into two fission fragments + 2–3 neutrons + energy, with an arrow labelled 'chain reaction possible'; (b) a FUSION PATHWAY — two hydrogen isotopes (deuterium + tritium, or two protons) fusing to form helium + a neutron + energy, labelled 'requires extreme heat/pressure'; (c) an $E = mc^2$ annotation linking both pathways, with the note 'mass defect → energy'; (d) an 'Applications' branch showing: nuclear power plant (KNEB), nuclear medicine (cancer treatment), and a warning symbol for nuclear weapons. At the bottom of their model, learners write a 2-sentence 'Model Update Statement' explaining what today's lesson added to their understanding of the driving question.	am going to draw the fission pathway on the board. Watch first, then draw it in your own model in your own style — it does not need to look like mine. What matters is that it shows the KEY ideas.' [Draw fission pathway on the board while narrating: 'One neutron hits U-235... it splits... we get two smaller nuclei, two or three more neutrons, and energy. Those neutrons can trigger MORE fissions — chain reaction.'] 3. Say: 'Now add the fusion pathway. You have seen this in the video. Two light nuclei... forced together under extreme heat... they merge... and release even MORE energy per kilogram than fission. Where does that energy come from?' [WAIT 10 seconds.] Cold-call 2 learners. Expected: mass defect / $E = mc^2$. 4. Say: 'Add your $E = mc^2$ annotation — link it with an arrow to BOTH fission and fusion. And add your applications branch.' 5. After 8 minutes of model building: 'Write your two-sentence Model Update Statement now. It must answer: What does today's lesson tell us about how we could HARNESS the energy in the pitchblende rock? You have 2 minutes.' [Allow 2 minutes.] 6. Cold-call 3 learners to read their Model Update Statement. After each, ask the class: 'Does this statement answer our driving question? Give me a thumbs up if it does.' Use thumbs as a quick class-level check. Revise any statement collaboratively if needed. 7. Close the lesson: 'Our model started in Lesson 1 with a rock fogging a film. Look at what it	Model Update Statement forces learners to explicitly articulate how new evidence changes their explanation of the anchoring phenomenon, practising the NGSS Science and Engineering Practice of 'Developing and Using Models' and 'Constructing Explanations.'	triggers a targeted prompt: 'What happened to the neutrons after the U-235 split? Add that.' (b) Model Update Statement — the 3 cold-called statements are assessed against this criterion: does the statement explicitly connect today's learning to the anchoring phenomenon (pitchblende rock / uranium) AND answer the 'harness' part of the driving question? Statements that do not mention uranium, energy, or harnessing are flagged for written feedback. (c) Thumbs check after each statement provides a rapid whole-class comprehension signal — if fewer than 70% of thumbs are up after the third statement, the teacher spends 2 additional minutes revisiting the key connection: 'The same instability that makes the rock emit radiation slowly — we can USE that same nuclear energy by triggering fission deliberately in a reactor.'

		shows now — natural decay, half-life, radiation types, detection, AND two ways humans have learned to deliberately release nuclear energy. In our final lesson, Lesson 7, we will complete the model by looking at how all of this is used — and misused — in medicine, agriculture, and energy in Kenya. But today, you answered the oldest question on our Driving Question Board: where does the energy come from?'		

D. TEACHER REFLECTION

After delivering this lesson, reflect on the following questions in your professional journal:\n\n1. SIMULATION ACCESS: Did all learner pairs have adequate access to the PhET simulation? If devices were limited, which parts of the simulation evidence were learners unable to gather — and how did this affect their comparison tables and model building? What offline alternative (e.g., a printed chain reaction card activity) could substitute in future?\n\n2. DEBATE QUALITY: Did learners cite numerical evidence from the evidence cards during the Structured Academic Controversy, or did they default to general/opinion-based arguments? If the debate became opinion-driven, consider pre-teaching the skill of 'evidence citation' in the next lesson by modelling an evidence-based claim explicitly.\n\n3. MASS DEFECT UNDERSTANDING: The ugali analogy is designed to make $E = mc^2$ accessible without formal mathematics. However, some Grade 10 learners may be ready for the equation and find the analogy insufficient. Did you observe learners asking for the formula? Note their names — they may benefit from an extension task in Lesson 7.\n\n4. KENYA CONTEXT ENGAGEMENT: Did the KNEB context (Kenya's real nuclear energy planning) motivate learners during the debate? Were learners aware of Olkaria geothermal as a comparison point? If not, consider adding a 5-minute 'Kenya Energy Mix' infographic at the start of the debate in future iterations.\n\n5. DQB PROGRESSION: Review the yellow sticky notes from today's DQB update. Which questions are answerable in Lesson 7 (applications: medicine, agriculture, radiation safety)? Which questions go beyond the curriculum (e.g., nuclear fusion reactors, ITER project)? Plan to explicitly acknowledge the 'beyond curriculum' questions in Lesson 7 as signs of genuine scientific curiosity — and point learners toward resources for further exploration.\n\n6. CUMULATIVE MODEL: Compare today's models against Lesson 5 models. Are learners adding the new content in a way that is CONNECTED to prior content, or are they drawing separate disconnected diagrams? If models are fragmented, dedicate the first 5 minutes of Lesson 7 to a 'model integration' activity where pairs compare models and add one missing connection.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Using the PhET Nuclear Fission simulation, learners observed: a U-235 nucleus splitting into two fission fragments + 2–3 neutrons + energy when struck by a neutron; a chain reaction branching diagram showing exponential growth from one initial neutron; the role of control rods in moderating the chain reaction; and the full energy transformation pathway inside a nuclear reactor (nuclear → thermal → kinetic → electrical energy). From the video, learners also observed a conceptual explanation of nuclear fusion and the KNEB context for Kenya's nuclear energy plans.
What did I learn?	Learners learned that: (1) Nuclear fission is the splitting of a heavy nucleus (e.g., U-235) by a neutron, producing fission fragments, more neutrons, and large amounts of energy; (2) Nuclear fusion is the joining of light nuclei (e.g., hydrogen isotopes) under extreme conditions, releasing even greater energy per kilogram; (3) In both reactions, the products have slightly less mass than the reactants — this 'missing' mass (mass defect) is converted to energy via $E = mc^2$, explaining WHERE the energy in the pitchblende rock ultimately comes from; (4) A controlled chain reaction in a nuclear reactor generates electricity, while an uncontrolled chain

	reaction produces a nuclear explosion — control rods manage this; (5) Kenya (through KNEB) is actively considering nuclear power as part of its energy mix alongside geothermal (Olkaria), hydro, and solar sources.
How does this explain the phenomenon?	Learners constructed evidence-based explanations that: (1) the energy released when our pitchblende rock emits an alpha or gamma ray comes from the conversion of a tiny amount of nuclear mass into energy (mass defect → $E = mc^2$), meaning the source of the radiation is literally the mass of the uranium nucleus itself; (2) nuclear fission and fusion are engineered applications of the same nuclear energy principle — deliberately triggering nuclear rearrangement to release large amounts of energy rapidly; (3) Kenya's decision about nuclear power involves weighing the benefit of large-scale, low-carbon electricity generation against risks of nuclear waste, safety incidents, and high upfront costs — a decision that requires scientific literacy, not just opinion.

LESSON 7 (80 minutes): Applications, Safety, and Unit Review: Closing the Case of the Invisible Killer	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners synthesise all prior learning across the unit by exploring real-world applications of radioactivity in Kenya (medicine, agriculture, industry, archaeology), evaluating radiation safety principles, presenting their final revised cumulative models, formally closing the Driving Question Board, and conducting a structured unit review.
Knowledge	Learners demonstrate understanding of: (a) applications of radioactivity including radiotherapy at Kenyatta National Hospital (KNH), KALRO crop irradiation for pest control and mutation breeding, industrial thickness gauging, and carbon-14 dating of Kenyan archaeological sites; (b) radiation safety principles including the ALARA principle (As Low As Reasonably Achievable), shielding materials (lead, concrete, water), the inverse square law of distance, and time minimisation; (c) the complete explanatory framework connecting isotope instability, decay types (α, β, γ), nuclear equations, half-life, fission, and fusion back to the anchoring phenomenon of pitchblende spontaneously exposing photographic film.
Skills	Learners are able to: (a) evaluate the benefits and risks of radioactive applications in Kenyan contexts; (b) apply the ALARA principle to design a radiation safety plan; (c) construct and present a final cumulative explanatory model comparing their Lesson 1 initial model to their final revised model; (d) use evidence gathered across the unit to formally answer the driving question; (e) close the Driving Question Board by categorising all posted questions as answered, partially answered, or requiring further study.
Attitudes	Learners appreciate the dual nature of radioactivity — its life-saving applications and its potential hazards — and demonstrate responsible, evidence-based decision-making when evaluating nuclear technologies in the Kenyan context. Learners value the collaborative process of model revision as a reflection of how scientific understanding grows over time.
Key Inquiry Question	How does knowledge of radioactivity help us solve problems in medicine, agriculture, energy, and environmental monitoring in Kenya — and how do we protect ourselves while harnessing its power?
Purpose in Storyline	This is the capstone lesson of the unit. Having spent six lessons building up the nuclear physics explanatory framework piece by piece — isotopes, decay types, nuclear equations, half-life, fission and fusion — learners now deploy that full framework to answer the driving question completely and formally. The pitchblende rock that puzzled learners in Lesson 1 is now fully explained: uranium-238 nuclei have a neutron-to-proton ratio outside the belt of stability, causing spontaneous alpha and beta decay chains that release penetrating radiation, continue indefinitely due to the long half-life of U-238 (4.5 billion years), and can be detected, harnessed, and safely managed using the physics principles learned throughout the unit. The DQB is closed, the cumulative model reaches its final form, and learners leave with both conceptual mastery and a sense of the societal relevance of nuclear physics in Kenya.
Safety Notes	No radioactive materials are used in this lesson. If printed images of radiation warning symbols (trefoil) or hospital/industrial photographs are used, ensure they are clearly labelled as illustrative only. When discussing medical and industrial applications, remind learners that licensed professionals operate under strict regulatory frameworks (Kenya Nuclear Regulatory Authority — KNRA). Emphasise that the ALARA principle is a professional and ethical obligation, not just a guideline.

B. LESSON OVERVIEW

Lesson 7 is the final, integrative capstone of the Radioactivity and Stability of Isotopes unit. The lesson opens by returning to the original pitchblende phenomenon — the rock that exposed photographic film in the dark — and challenging learners to now explain it completely using everything they have learned. Through a structured applications carousel (radiotherapy at KNH, KALRO crop irradiation, industrial thickness gauging, carbon-14 dating of Kenyan fossils), learners connect nuclear physics

concepts to Kenyan real-world problems. A radiation safety design task applies the ALARA principle. Groups then present their final cumulative models — revised from their Lesson 1 naive sketches — narrating the evolution of their thinking. The Driving Question Board is formally closed as a whole-class ceremony: each posted question is read aloud and the class decides whether it is fully answered, partially answered, or flagged for further study. A structured concept-map review consolidates all key learning outcomes. The lesson closes with learners writing a final individual 'Letter to a Year 9 Student' explaining why some rocks are radioactive and why that matters for Kenya — a summative writing task that doubles as a transfer assessment.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Chronometric revolution Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Radioactivity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown the original Lesson 1 image: the pitchblende rock next to the exposed photographic film in complete darkness. They are given 3 minutes of silent individual thinking time to write a FINAL COMPLETE EXPLANATION of the phenomenon in their science notebooks — using as many unit concepts as possible. They then share with a shoulder partner for 2 minutes, comparing their explanation to their Lesson 1 initial prediction (which they retrieve from their notebooks). Learners annotate: 'What did I think then? What do I know now? What changed?'	Display the original phenomenon image prominently. Say verbatim: 'You have seen this rock before — on the very first day of this unit. You wrote your best guess about what was happening. Today, I want you to write your FINAL, COMPLETE, EVIDENCE-BASED explanation. You have seven lessons of evidence. Use all of it. You have 3 minutes of silence.' Set a visible timer. After 3 minutes, say: 'Now compare your explanation today with what you wrote in Lesson 1. What is different? What surprised you about how much your thinking has changed?' Allow 2 minutes of partner talk. Cold-call 3 students — choose one high-performer, one mid-level, one who has grown visibly during the unit — and ask: 'What is the single most important idea that changed your thinking?' Apply 10-second WAIT TIME before accepting responses. Record key phrases on the board under the heading: 'Our Final Explanation — Building Blocks.'	Reflective Comparison ('Then vs. Now'): By explicitly retrieving and comparing their Lesson 1 naive model with their current understanding, learners consolidate the arc of conceptual change across the unit. This metacognitive strategy — a form of self-explanation — strengthens long-term retention by making learning visible to the learner themselves. It also primes engagement with the final model-building phase.	Observable indicator: Learners' written explanations should now include at minimum — isotope instability (neutron-to-proton ratio), identification of decay type (alpha for U-238), a nuclear equation, reference to half-life (why the rock 'never stops'), and penetrating radiation reaching the film. Teacher circulates during silent writing and uses a rapid 3-point mental tally: (1) mentions nuclear instability, (2) mentions correct decay type, (3) mentions half-life. Any learner missing two or more of these is flagged for targeted support during the applications carousel phase.
Observe Phase 📹 **VIDEO:** Chronometric revolution Source: Khan Academy (English	Four stations are set up around the classroom, each with a printed information card, a photograph, and a task card. Learner groups (4–5 per group) rotate every 4 minutes. Station 1 — KNH	Before rotation begins, say: 'Each station is a real application of the nuclear physics we have studied. Your job is not just to read — it is to CONNECT each application back to our original rock and our	Application Transfer Carousel: By encountering the same core concepts (decay types, half-life, penetrating power, nuclear equations) deployed across four distinct Kenyan real-world	Observable indicator: Station task cards are collected at the end of the carousel. Teacher scans for three key errors: (1) At Station 3, learners incorrectly choosing alpha or gamma instead of beta for

- US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Radioactivity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Radiotherapy: A case study of a Nairobi patient receiving cobalt-60 (Co-60) radiotherapy for cervical cancer (Kenya's most common cancer in women). Task: Identify the isotope, write its nuclear symbol, explain why gamma radiation is used for treatment, and state one safety measure for the radiographer. Station 2 — KALRO Crop Irradiation: A description of how the Kenya Agricultural and Livestock Research Organisation uses gamma irradiation to sterilise pests in stored maize and beans and to induce beneficial mutations in sorghum varieties for semi-arid counties like Makueni. Task: Explain how radiation affects living cells, identify the benefit vs. risk trade-off, and link to half-life. Station 3 — Industrial Thickness Gauging: A diagram of a paper or aluminium sheet manufacturing plant using a beta-source detector to measure sheet thickness continuously. Task: Explain why beta particles (not alpha or gamma) are used, and how the detector reading relates to thickness. Station 4 — Carbon-14 Dating: A photograph of the Homo habilis fossil discovered at Koobi Fora, Lake Turkana, Kenya. Task: Use the half-life of C-14 (5,730 years) to explain why C-14 dating cannot be used for this fossil (too old), and identify which isotope system (uranium-lead) would be more appropriate and why.	driving question. Every concept we have studied lives inside these four stations.' Distribute station cards. Set a 4-minute rotation timer with an audible signal. Circulate continuously — spend approximately 1 minute per station per rotation cycle. At Station 4, prompt with: 'Why can't we use carbon-14 on a 1.8-million-year-old fossil? What does that tell you about the relationship between half-life and the age of the sample?' Apply 12-second WAIT TIME. At Station 1, prompt: 'Why is the radiographer standing behind a lead screen? What property of gamma radiation makes that necessary?' Cold-call 2 students at the end of the carousel with: 'Tell me one thing from your station that connects directly back to the pitchblende rock in Lesson 1.'	contexts, learners deepen understanding through contextual variation. Each new context requires learners to identify WHICH concept applies and WHY — a higher-order transfer task that consolidates understanding more robustly than re-reading notes. The Koobi Fora fossil station is deliberately chosen to create productive confusion (C-14 doesn't work here) — this 'desirable difficulty' strengthens the learner's ability to reason about the limits of nuclear techniques.	thickness gauging — indicates incomplete understanding of penetrating power; (2) At Station 4, learners attempting to use C-14 for the Homo habilis fossil — indicates incomplete understanding of half-life range applicability; (3) At Station 1, learners unable to explain why gamma is used therapeutically — indicates disconnect between penetrating power and biological application. These errors are addressed in the Explain phase.
Explain Phase 📹 **VIDEO:** Chronometric revolution Source: Khan Academy (English - US curriculum)	Part A — Radiation Safety Design (10 min): Each group receives a scenario card: 'You are advising the Kenya Nuclear Regulatory Authority (KNRA) on safety protocols for a new nuclear medicine unit at Moi Teaching and	For Part A, distribute the KNRA scenario card and say: 'The ALARA principle is not just a physics idea — it is a professional and ethical obligation for everyone who works with radioactive materials in Kenya. Your safety	Design-Based Sensemaking (Safety Plan as Model Application): Designing a safety protocol requires learners to deploy knowledge as a tool for solving a real problem — the highest level of transfer. By	Observable indicator: In the safety plan, learners should: (a) correctly identify lead or concrete as shielding for gamma-emitting I-131; (b) state that doubling distance reduces dose to one-quarter (inverse square law); (c)

🔍 Search ARES for similar videos 📖 **HTML:** Radioactivity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Referral Hospital in Eldoret. The unit will use iodine-131 (I-131, half-life 8 days) for thyroid cancer treatment. Design a safety plan for (a) the patient, (b) the medical staff, and (c) the hospital waste disposal team, using the ALARA principle — As Low As Reasonably Achievable.' Groups must reference shielding, distance (inverse square law), time limitation, and half-life calculations in their plan. Part B — Final Explanation Synthesis (10 min): Each group returns to their cumulative model poster (carried through all 7 lessons). They make any final additions — specifically: applications arrows showing how the same nuclear instability that exposed the pitchblende film is now being harnessed in hospitals, farms, and factories — and prepare a 2-minute group presentation narrative.	plan must be specific: name the shielding material, calculate what happens to the dose when you double the distance, and use the I-131 half-life to advise when waste is safe for disposal.' Circulate and challenge each group with: 'How many half-lives must pass before the I-131 waste activity drops to less than 1% of its original level? Show the calculation.' (Answer: approximately 7 half-lives = 56 days.) Apply 15-second WAIT TIME after posing this question. For Part B, say: 'Your model started as a sketch of a puzzling rock. Today it should be a complete explanatory framework for nuclear physics. Make sure your model tells a STORY — from unstable nucleus to emitted radiation to real-world application.' Cold-call 2 groups to preview one sentence of their narrative before formal presentations.	anchoring the design task in a recognisable Kenyan institution (MTRH Eldoret) and a specific isotope (I-131) with a short half-life, learners must integrate penetrating power, inverse square law, ALARA, and half-life decay simultaneously. This integrative application reveals gaps in understanding (particularly around half-life calculation and shielding material selection) while consolidating connections between concepts. The Final Explanation construction directly addresses the NGSS practice of 'Constructing Explanations' — learners articulate a causal mechanism, not just a description.	calculate approximately 56 days (7 half-lives) for waste to fall below 1% activity. In the model, learners should explicitly label an 'applications' layer showing the link from nuclear instability → controlled radiation use → societal benefit. Any group unable to perform the half-life calculation is given a worked example prompt card by the teacher before presentations begin.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Chronometric revolution Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Radioactivity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The full Driving Question Board — maintained across all 7 lessons — is displayed at the front of the room. The teacher reads the original driving question aloud: 'Why do some atoms in Kenyan rocks spontaneously release invisible energy — and how can we harness or protect ourselves from that energy?' Each group then presents their 2-minute final model narrative, connecting at least three unit concepts back to the pitchblende phenomenon. After all presentations, the class conducts a formal DQB Closure Ceremony: the teacher reads each posted question card aloud, and learners vote (show of hands or green/yellow/red card signals) to classify it as: GREEN — fully answered by our unit, YELLOW —	Begin by saying: 'Seven lessons ago, this board was empty and this rock was a mystery. Today, we close the board. But first — your models.' Allow each group 2 minutes strictly (use a timer). After presentations, move to the DQB Closure Ceremony. For each card, read it aloud and say: 'Green, yellow, or red? Show me your card.' After voting, ask one student to justify the classification with: 'Tell me in one sentence what evidence from our lessons answers this question.' Apply 10-second WAIT TIME. After all cards are classified, say: 'Now — the driving question itself. I am going to write our class answer on the board. I need four volunteers to each contribute one sentence.' Cold-call 4 students who have not	Consensus Model Building + DQB Closure: The Driving Question Board closure is a structured epistemic ritual that makes the growth of collective knowledge visible. By classifying each question as answered, partial, or open, learners practise the scientific norm of calibrated confidence — distinguishing what is known from what remains uncertain. The group model presentations enact the NGSS practice of 'Obtaining, Evaluating, and Communicating Information,' requiring learners to translate their cumulative understanding into a coherent public narrative. The collaborative construction of the 4–5 sentence consensus answer to the driving question is a high-leverage sensemaking move: it	Observable indicator: The consensus driving-question answer on the board should include all five required elements (unstable nucleus, decay type, penetrating radiation, half-life, applications/safety). If any element is missing after four student contributions, the teacher prompts with a targeted question: e.g., 'We have described the radiation — but why does the rock NEVER STOP emitting? What concept explains that?' Individual group model presentations are assessed against a 3-point rubric (displayed on the board): 1 = mentions decay type only; 2 = connects decay type + half-life + one application; 3 = full causal chain from nuclear instability through all major unit concepts to Kenyan application

	partially answered, more evidence needed, RED — beyond our current study, flagged for further investigation. A volunteer scribe records the classification on each card. The driving question itself is answered collaboratively as a class, with the teacher scribing a 4–5 sentence consensus answer on the board.	yet spoken today. The final written answer must include: unstable nucleus → decay type → penetrating radiation → half-life → applications and safety. End by saying: 'The rock is no longer a mystery. But notice how many yellow and red cards remain. That is not failure — that is the frontier of science. Kenyan scientists at the Kenya Nuclear Regulatory Authority are working on those very questions right now.'	forces the class to agree on a shared, evidence-based explanation rather than holding parallel private understandings.	and safety. Teacher records each group's score for formative tracking.
Model Building Phase 📹 **VIDEO:** Chronometric revolution Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Radioactivity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Final model posters are displayed gallery-style around the room for a 3-minute silent gallery walk — learners place one green sticky note ('I see this concept clearly') and one orange sticky note ('I want to ask about this') on any poster that is not their own. Learners then return to their seats for the individual summative transfer task: 'Write a Letter to a Year 9 Student (150–200 words) explaining: (1) Why some rocks in Kenya — like the pitchblende found in certain geological formations — spontaneously release invisible energy; (2) What that energy is and why it is both dangerous and useful; (3) One specific way Kenya is using or managing this phenomenon for the benefit of its people.' Learners write individually for 10 minutes. The letter is handed in as an exit artefact. In the final 2 minutes, the teacher leads a rapid whole-class concept map recitation: the teacher points to unlabelled nodes on a projected concept map of the unit and cold-calls students to name the concept and state one connection.	For the gallery walk, say: 'You have 3 minutes. Read at least two other groups' models. Leave a green note where you see something your own model also shows. Leave an orange note where you have a question or see something different from your own model. Be specific — write a full sentence, not just a word.' Set a 3-minute timer. For the letter task, say: 'This letter is your individual final explanation. No group help. Your Year 9 reader has never studied radioactivity. Use clear language, correct scientific terms, and connect everything back to the Kenyan context. You have 10 minutes.' Circulate silently during writing. In the final 2 minutes, project the concept map and say: 'Fast round — I point, you name and connect. No hands — I will cold-call.' Cold-call 6 students rapidly across the concept map nodes. End the lesson by returning to the original image of the pitchblende rock and saying: 'This rock is no longer invisible to you. You can see inside it — all the way to the nucleus. That is what physics does. It makes the invisible visible.' Pause 5 seconds. 'Well done.'	Transfer Writing (Letter Task) as Final Model Externalisation: The 'Letter to a Year 9 Student' task is a high-transfer writing activity that requires learners to translate technical understanding into accessible language for a novice audience — a demanding cognitive task that reveals the depth and coherence of their understanding. Research consistently shows that explaining to others (real or imagined) is one of the most effective consolidation strategies. The gallery walk preceding the letter primes recall of the unit's conceptual map by activating visual memories of the cumulative model. The rapid concept-map recitation in the final 2 minutes serves as a whole-class spaced retrieval practice session, reinforcing key vocabulary and connections.	Observable indicator: Letters are assessed against three criteria: (1) Scientific accuracy — correct identification of nuclear instability, decay type, and half-life as the cause of the phenomenon; (2) Kenyan relevance — explicit mention of a Kenyan application (KNH, KALRO, KNRA, Koobi Fora, etc.); (3) Dual nature — acknowledgement of both the hazard and the benefit of radioactivity. Letters meeting all three criteria are rated 'Meeting Expectations.' Letters meeting two criteria are 'Approaching Expectations' and are targeted for written teacher feedback. Letters meeting one or zero criteria are flagged for individual follow-up before the unit assessment. Teacher also notes any concept-map recitation errors during the rapid final round and records them as indicators for revision before the formal unit test.

D. TEACHER REFLECTION

After delivering this final lesson, reflect on the following questions in your professional journal:

1. DRIVING QUESTION CLOSURE: When the class constructed the consensus answer to the driving question on the board, was the answer scientifically complete? Did learners spontaneously include all five required elements (nuclear instability, decay type, penetrating radiation, half-life, applications/safety), or did you need to prompt heavily? What does this tell you about which concepts need reinforcement before the formal unit assessment?

2. MODEL EVOLUTION: Compare the Lesson 1 naive models with the final models displayed in the gallery walk. Identify the single conceptual leap that was most difficult for learners across the unit. Was it the neutron-to-proton ratio and the belt of stability? The distinction between fission/fusion and radioactive decay? The inverse square law? Plan one targeted retrieval activity on this concept for the start of the next unit.

3. CAROUSEL STATION ERRORS: Review the collected Station 4 (Carbon-14 / Koobi Fora) task cards. How many groups incorrectly tried to apply C-14 to the Homo habilis fossil? If more than one-third of groups made this error, the relationship between half-life range and dating applicability requires explicit re-teaching in the unit review lesson or homework task before the assessment.

4. DQB YELLOW AND RED CARDS: Which questions remained yellow or red at the end of the DQB closure ceremony? These unanswered questions are valuable: they can be used as extension research tasks, as prompts for the Kenya Science Fair, or as hooks for the next physics unit. Consider posting them in the classroom as a 'Questions We Are Still Investigating' wall.

5. LETTER TASK AS ASSESSMENT DATA: Read all letters before the next lesson. Calculate the percentage of learners who: (a) correctly identified nuclear instability as the root cause; (b) included a Kenyan application; (c) acknowledged both hazard and benefit. Use this data to inform the structure of your unit revision session and to identify learners who may need additional support before sitting the formal unit assessment. Share anonymised high-quality letters (with learner permission) as mentor texts for the next cohort.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed four real-world Kenyan applications of radioactivity through the applications carousel: cobalt-60 radiotherapy at Kenyatta National Hospital for cervical cancer treatment; KALRO gamma irradiation of stored maize, beans, and sorghum for pest control and crop improvement in counties like Makueni; beta-source industrial thickness gauging in manufacturing; and carbon-14 versus uranium-lead dating applied to the Homo habilis fossil at Koobi Fora, Lake Turkana. Learners also observed their own Lesson 1 naive models alongside their final cumulative models, making the arc of conceptual change visible.
What did I learn?	Learners learned that the same nuclear instability that caused the pitchblende rock to expose photographic film in Lesson 1 is the foundational phenomenon behind life-saving medical treatments, improved crop yields for Kenyan farmers, precise industrial quality control, and the dating of Kenya's extraordinary hominin fossil record. They learned that radiation safety is governed by the ALARA principle — minimising exposure through shielding (lead, concrete), maximising distance (inverse square law: doubling distance reduces dose to one-quarter), and minimising time — and that the half-life of an isotope determines both its usefulness and the duration of its hazard. They consolidated the complete unit explanatory framework: unstable nucleus (imbalanced neutron-to-proton ratio) → spontaneous decay (α, β, or γ) → nuclear equation → half-life → penetrating radiation → detectable, harness-able, and manageable energy.

How does this explain the phenomenon?	Learners explained, in their own words and in writing, why the pitchblende rock spontaneously and continuously exposes photographic film without any external energy input: uranium-238 nuclei have a neutron-to-proton ratio outside the belt of stability, causing spontaneous alpha decay (and a subsequent decay chain including beta and gamma emissions), releasing penetrating radiation that passes through the film container. The process continues indefinitely on human timescales because U-238 has a half-life of approximately 4.5 billion years, meaning only an infinitesimally small fraction of nuclei decay in any observable period — yet the number of atoms in even a small ore sample is so enormous (on the order of 10^{23}) that the count rate remains measurable and persistent. Learners also explained that this same physics, properly understood and controlled, enables Kenya to treat cancer patients, protect stored food, ensure product quality, and read the deep history of human origins from the rocks of the Rift Valley.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.